BECKMAN COULTER BANGALORE DEVELOPMENT CENTRE - REQUEST FOR PROPOSAL – SUPPLY INSTALLATION TESTING AND COMMSISIONING OF DOCK LEVELLER

CBRE SOUTH ASIA PVT LTD
PROJECT MANAGEMENT, INDIA

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1. CHECKLIST FOR DOCUMENTS TO BE ENCLOSED WITH THE BID

The below listed documents to be submitted along with the sealed bid documents, any document that is not enclosed will term the bid insufficient and may not be considered for further process.

1	Tender Document as issued – every page duly filled, sealed, signed / initialed	Yes / No
2	Tender drawings issued – sealed, signed / initialed	Yes / No
3	Letter indicating any deviation / assumption / conditions, if any	Yes / No
4	Form of Tender filled up	Yes / No
5	Details of Plant, Machinery, Equipment & Personnel proposed to be deployed for the project	Yes / No
6	Resume / CV of the proposed deployment	Yes / No
7	Details of Management / Technical Team to be deployed.	Yes / No
8	Site Organization Chart specific to the project	Yes / No
9	Time Schedule	Yes / No
10	Construction Methodology	Yes / No
11	Quality Assurance & Quality Control Plan	Yes / No
12	Safety, Environment & Occupational Health Plan	Yes / No
13	Bill of Quantities" with rates and amount duly filled in	Yes / No
14	Income Tax Returns for last 3 Financial Years	Yes / No
15	ESI & PF Registration certificate	Yes / No
16	GST and other Taxes Registration & Certificate as applicable	Yes / No
17	PAN Number & Copy of PAN	Yes / No

2. NOTICE INVITING TENDER

A. Beckman Coulter (CLIENT) having its Registered Office at 50-B, II Phase, Peenya, Near TVS Cross, Peenya Industrial Area, Bangalore, Bengaluru (Bangalore) Urban, Karnataka, 560058, India (hereinafter referred to as the "Client") invite limited tenders from the pre-qualified Vendors for Dock Leveller -Supply and installation at Beckman Coulter BDC (Hereinafter referred to as the Project), KIADB, Mahadeva Kodigehalli, Bengaluru, Karnataka, 562149, India (Hereafter referred to as "The Site").

B. CBRE South Asia (P) Ltd (CBRE) having their Regional Office at Ground Floor, The, Hulkul Brigade Centre, No 82, Lavelle Road, Shanthala Nagar, Ashok Nagar, Bengaluru, Karnataka 560001, India (hereinafter referred to as the "Project Manager") have been appointed by CLIENT, to act as the Principal Contractor for the Project.

1. The tender documents comprise of two parts – Technical Bid & Commercial Bid to be submitted separately in sealed envelopes. The priced bill of quantities and covering letter will constitute the commercial bid rest all documents like drawings, write up, specifications etc. shall be termed under technical bid. A signed copy of this RFP/tender along with the forms enclosed with shall be returned with the tender towards technical bid.
2. Tender Documents will be issued to the shortlisted bidders through **Email. The tender documents shall be** duly printed, sealed and signed on all pages by the bidders, for submission at Client office (Beckman Coulter - 50-B, II Phase, Peenya, Near TVS Cross, Peenya Industrial Area, Bangalore, Bengaluru (Bangalore) Urban, Karnataka, 560058, India) to **Kind Attn. of Mr. Arun Kumar / Mr. Sameer Khan** on or before **11/3/2020 by 16:00 Hours**
3. Tenders that are not in accordance with this notification will not be entertained.

4. The Client reserves the right to extend the period, to reject any or all the tenders and/or to accept the lowest or any other tender, in whole or in part, without assigning any reason whatsoever thereof.
5. **Period of Completion**
    a. The entire scope of Package works shall be as per tender, to be completed in all respects within discussed timelines. The issue of Work Order will proceed in a manner in accordance with the schedule laid down in this document.

Project Milestone	
Shop drawing /Mobilization	3 days from the date of Email confirmation
Supply of approved Materials / Equipment's	Within 4 weeks from Date of Shop drawing / Schematic approval
On site development	2 Weeks from the date of Work front clearance
Final hand over	3 days from the date of Work completion & Virtual hand over

    b. The tenderer should examine carefully the accompanying tender documents and obtain all the information including inspection at site at his cost, prior to submission of his tender. Tenderer are deemed to have inspected the site and to have made all investigations regarding the nature, extent and scope of work to be executed including the site conditions. No claim of any kind shall be entertained at any time, on account of lack of knowledge of the details thereof, such as: climatic, law and order, labor conditions, site conditions, availing of amenities or facilities, etc.
6. The tenderer shall quote all-inclusive rates. The rate shall include all applicable taxes, levies, chess, duties, etc., as prescribed by the Govt/Statutory bodies. There shall be no extras on any account whatsoever, excluding the statutory variations, if any, that may accrue after award of contract. The sum payable shall be calculated on the basis of the unit prices specified in the contract and of the work actually executed in accordance with Drawings, technical specification and Bill of Quantities. Taxes, as applicable on the transaction, shall be calculated and shown separately in the Price schedule. The Quote shall show the tax components

applicable separately from the base price. If the price quoted is inclusive of all taxes, the percentage and amount of tax considered shall be mentioned.

7. This tender, together with the final letter of offer by Vendor after due negotiation and with the Client's written acceptance by way of an Agreement or Work Order, any other agreed conditions / documents, including but not limited to the drawings, specifications, bill of quantities, appendices etc., shall constitute a binding contract between the Client and the Vendor for the execution of the Contract.
8. Upon acceptance of the tender, the rates quoted shall be fixed, firm and final and shall remain valid for the complete duration of the contract and till completion of project and no escalation in rates shall be permitted for any reason whatsoever.
9. The Client reserves the right to make changes, by giving in writing, to the terms and conditions of the contract and the amended terms and conditions shall supersede any corresponding terms and conditions in the existing contract and shall so bind the parties.
10. The Tenderer agrees that any communication made by it to the Client or the Project Managers, which relates to the terms and conditions or in any manner pertaining to the contract, shall not bind the Client or the Project Managers unless the same is incorporated in the terms and conditions and signed by the parties.
11. The tender shall remain valid and open for acceptance for a period of 90 days from the specified date of receipt of tender and during this period, the tenderers shall not be entitled to withdraw or cancel the tender submitted by it/him. This can be extended, if necessary, only with the written consent of the tenderer.
12. Earnest Money Deposit (EMD)
    a. The tenderer shall, along with its/his tender, submit Earnest Money Deposit of value 1% of the contract value in the form of Bank Guarantee/Demand Draft from an approved scheduled bank in the format indicated in the Appendix – D in favor of Client. The Bank Guarantee shall be valid for a period of 90 Days from date of submission of the tender. The Earnest Money Deposit may be forfeited for the following reasons and the same shall be

without prejudice to all other rights of Client that it may have for recovering any damages or losses incurred or suffered by it by it in this regard:

i. If the tenderer withdraws its tender after tender submission, except where written notice of the modification or withdrawal is received by the Client prior to the deadline for submission of tenders.
ii. In the case of a successful tenderer, if it fails within the specified time limit to: i) Sign the Contract Agreement, ii) Furnish the required performance security

13. The tenderer is required, at his expense, to obtain all the information he may require enabling him to submit his tender including necessary visits to the site, procurement of necessary materials, labor, etc., and is required to comply with the requirements of the local/government/public authorities in such matters.
14. A schedule of approximate quantities for various items accompanies this tender. The Client do not accept any responsibility for the correctness or completeness of this schedule in respect of items and quantities and is liable to alteration by omissions, deductions, or additions at the discretion of the Client without affecting the terms of contract. The tenderers should satisfy themselves about the completeness and accuracy of the same before submitting the tender.
15. Clarifications if any, regarding tender documents shall be directed to **Kind Attn. of Mr. Arun Kumar Rajendran / Mr. Sameer Khan on or before 07/3/2020 by 16:00 Hours** and the clarifications will be circulated to all the tenderers by mail.
16. No claims about for want of information of any point or any change in rate or conditions after the opening of tender shall be entertained.
17. Only those tenders fully completed in writing on the form of tender together with all the documents and received by the time and date specified hereunder will be considered. (All copies of drawings if provided should be returned unmarked). The intellectual property in the drawings belongs exclusively to the Client. The tenderers are not entitled to use the same for any purpose whatsoever, under any circumstances, other than the fulfillment of the obligations under this tender.
18. All information supporting the tender shall be in English and all entries made by hand shall be written in ink. There shall be no over-writing or erasure. All corrections

should be attested by the tenderer with its/his dated initials as many times as the corrections occur.

19. The tenderer should fill in the rates tendered in figures as well as in words. The amount for each item should be worked out and the requisite totals indicated.
20. If on checking, differences are found between the rates given by the tenderer in words and figures or in the amounts worked out by him the following procedure shall be followed for the tenderer to carry out necessary corrections: -
    a. Where there is difference between the rates in figures and in words, then the rate quoted by the tenderer in words shall be taken as correct.
    b. Where the amount of an item is not worked out by the tenderer or it does not correspond with the rate written either in figures or in words, the rates that correspond to the amounts worked out by the tenderer shall be taken as correct.
    c. Where the rates quoted by the tenderers in figures and in words tallies but the amount is not worked out correctly, the rate quoted by the tenderer shall be taken as correct and the amount corrected accordingly.
21. The tenderer is required to check the numbers of the pages and should any be found missing or in duplicate or the figures or writing indistinct, he must inform the Client and CBRE at once and have the same clarified.
22. The tender shall remain valid for acceptance for a period of 60 days counted from the day following the latest date fixed for receiving tenders. No notice shall be entertained regarding any change during this period unless mutually agreed to between the Client and the tenderer.
23. The tender sum and pricing of bill of quantities shall be given in the Indian currency.
24. The basic prices of processed raw materials and manufactured products must be indicated along with the relevant item of work described in Schedule of Quantities. The Vendor shall quote his rates based on the basic rates as indicated.
25. The tenderer whose tender is accepted will be required to enter into a Contract with the Client as per the Conditions of Contract attached herewith, which shall prevail. A binding relationship shall arise between the parties on the execution of such contract.

26. The tenderer is required to submit satisfactory evidence of his experience and that of the recently executed works.
27. The Client reserves the right to accept any tender in part or full or reject any tender without giving any reason whatsoever and does not bind himself to accept the lowest or any tender.
28. The tenderer has to read the drawings in conjunction with technical specifications and the Bill of Quantities to obtain all the necessary information for submission of the offer. In the event of any discrepancy between the details and or description given in the bill of quantities, the drawings, and the technical specifications, then the item shall be deemed to have been priced in accordance with the details and / or description given in the following order of precedence:
    1. Tender BOQ
    2. Drawings
29. The documents forming the contract are to be taken as mutually explanatory of one another. For the purpose of interpretation, the priority of the documents shall be in accordance with the following sequence:
    a. Contract Agreement
    b. Work / Purchase Order
    c. Letter of Acceptance
    d. Notice Inviting Tender & Conditions of Bid
    e. Appendix to Tender
    f. Special Conditions if any agreed between parties before execution of agreement
    g. General Conditions of Contract
    h. Bill of Quantities, Schedules & any other documents forming the part of the contract.
    i. Drawings
    j. Technical Specifications

    If an ambiguity or discrepancy is found in the documents, the Project Manager shall issue necessary clarifications or instructions.

30. In all the cases, it is understood that the details and/or description not specifically mentioned in the Bill of quantities and/or the drawings shall be the same as those mentioned in the technical specifications.
31. Client may also nominate the subcontractor at any stage and back to back terms and conditions will be followed with OHP of 2.5% on subcontractor PO value for main contractor.
32. Any further interpretation of above clauses shall be at the discretion of the Client / Project Managers, whose decision shall be final and binding on the Vendor.

GENERAL CONDITIONS

3.1 SCOPE OF WORK:

Supply ,Installation Testing and Commissioning of Dock Leveler as per the parameters given enclosed herewith and shall include the specifications, drawings and sketches issued to you and approved shop drawing for color scheme by our Project Managers.

3.2 CONTRACT PRICE:

a) The contract price would be as per the final agreed price via BOQ
b) The contract price shall be all inclusive of basic price, all applicable taxes, duties, levies and cess. You will be paid as per accepted rates as given in the enclosed Bill of Quantities.
c) All taxes shall be quoted separately and mentioned in the offer.

3.3 SPECIFICATION

The specifications for various works shall be as mentioned in the technical specification. This will be read in conjunction with Para 29 of NIT above.

3.4 DATE OF COMMENCEMENT:

The date of commencement of work shall be from the date of email confirmation/ issue of LOA / WO/PO (Work/Purchase Order) from CBRE, whichever is earlier.

3.5 DATE OF COMPLETION:

You shall complete the works in all respects & ready for usage and hand over to us As per the milestone mentioned in GCC. Please note that this period includes the days of inclement weather.

3.6 PAYMENT TERMS:

- 60% against delivery
- 25% against installation
- 10% against commissioning
- 5% of contract value would be retention payable after 24 months of DLP [Defect Liability Period] or against equivalent irrevocable bank guarantee valid for 24 months from date of substantial completion or date of satisfactory acceptance whichever is later

5% Performance Bank Guarantee of total contract value to be submitted separately valid from the date of PO till completion of the works as per the approved format. PBG shall be submitted along with Advance Bank Guarantee

3.7 PENALTY:

The penalty for delay shall be 1% of the total contract value per day of delay, not limited to project completion but each stage of work, subject to a maximum of 10% of the total contract value.

3.8 WARRANTY & GUARANTEE:

The supplied product shall be covered by warranties & guarantee in accordance with standard duration associated with the product. Any damage to consignment prior to hand over will be on account of the Vendor. Warranties & Guarantee, for a minimum of two year and a maximum as allowable for the product and / or described in spec sheet, shall be provided.

3.9 NOMINATED SUBCONTRACTOR:

Client may also nominate the subcontractor at any stages and back to back terms & conditions will be followed with OHP of 2.5% on subcontractor PO value to main contractor.

3.10 COMPLIANCE OF LABOUR REGULATIONS:

Vendor shall comply with all laws applicable to workmen employed/contracted by –him and the rules framed there under. The Client and other consultants as engaged by him on this project shall in no event and under no circumstances, be liable or responsible for any default by way of non-observance/compliance of the said law/rules on part of the Vendor and the Vendor shall further indemnify them against any liabilities and costs/expenses from all proceedings in respect thereof. Vendor further agrees that the Client reserves the right to obtain/procure any such statutory/legal compliances, as may be applicable to this project that the Vendor is expected to comply with but has not (due to any reason whatsoever), through any monies/guarantees available with the Client, at the Vendor's risk and cost.

3.11 SAFETY PRECAUTIONS AND PROGRAMS & SECURITY:

You shall be responsible for all the materials brought by you to the site and shall secure the same within the premises at your risk and cost.

You shall be responsible for initiating, maintaining and supervising all safety precautions and programs in connection with the Work.

3.12 SAFETY OF PERSONS AND PROPERTY

Vendor shall take all reasonable precautions for the safety of, and shall provide all necessary protection to prevent damage, injury or loss to:

1. All employees on the Work and all persons who may be affected thereby; All the work and all the materials and equipment to be incorporated therein, whether in storage on or off the site, under Vendor's care, custody or control or any of Vendor's Sub-Vendor or Sub-sub-Vendor; and

2. Other property at the site or adjacent there to, including trees, shrubs, laws, walks, pavements, roadways, structure and utilities not designated for removal, relocation or replacement in the course of construction/ installation.

3. Vendor shall give all notices and comply with all applicable laws, ordinances, rules, regulations and lawful orders of any public authority bearing on the safety of persons or property or their protection from damage, injury or loss.

4. Vendor shall promptly remedy all damage and loss to any property caused in whole or in part by him or any one directly/ indirectly employed by him.

5. Vendor shall not load or permit any part of work to be loaded so as to endanger its safety.
6. In addition to above Vendor shall undertake to comply with OHSE policy of Client.

3.13 EMERGENCIES

In any emergency affecting the safety of persons or property, Vendor shall act immediately, in keeping with the Statutory & Legal requirements and Industry practices, to prevent threatened damage, injury or loss but at the same time safeguarding the interest of the Clients and the

project Vendor shall comply with (including, but not limited to) the following towards emergency preparedness: -

a. providing adequate fire points and regularly maintain the record for maintenance,
b. Overhead cable management with ELCB's, & strictly avoid unsafe power tapings. Use cable connectors & avoid insulation tapes for cable joints, Using of double insulated cables, Industrial plugs & sockets etc.
c. Maintain access/egress free from obstructions by organizing good housekeeping at all-time.
d. Have special focus on effective waste management.
e. conduct safety awareness trainings on regular basis and maintain records
f. Nominate emergency response team.
g. have emergency transport (4 wheeler) at all times of executing the works
h. maintain basic first aid material

SPECIAL CONDITIONS

4.1 DELIVERY STATUS REPORTS

The vendor shall submit to CBRE by weekly Delivery Status Reports and stating the acknowledged shipping date, estimated warehouse arrival date, anticipated site delivery date and project delivery date.

4.2 DELIVERY AND STORAGE

a. The vendor shall upon receipt of all the material inspect them for damage and compliance with the Contract Documents.

b. The vendor shall be entirely responsible for the receiving, handling, storage, security and transportation for delivery and placing of furnishings including insured warehouse/Site store till handing over of Site.

c. The vendor shall store the received material only at a place approved by the Project Manager/ Client. The material will be stored such that it is not subjected to damages/wear and tear and all necessary precautions for the same shall be put in place. Packaging material for the imported items shall not be of wood and if it is then the vendor needed for fumigation shall accompany the shipment. If the material needs to be fumigated in India, then any delay & demurrages (if any) due to this will be vendor's account.

4.3 SUBSTITUTIONS

Performance of the contract shall be based upon materials and products specified in the Contract Document. No substitutions will be made by the vendor without the written approval of Client.

4.4 INSTALLATION

a. Installation and placement shall be based on the Drawings and Specifications.

b. It will be vendor's responsibility to organize all the tools & equipment for carrying out the installation & laying work

c. It is the responsibility of the Vendor for adhering delivery and installation schedules as coordinated by Project Manager/ client and give twenty-four-hour notice before scheduled installation. Any additional costs incurred by the Vendor due to lack of adherence to the schedule will be the vendors' responsibility.

4.5 CLEANING, REPAIR AND REMOVAL

a. Remove labels, paper, tape and wrapping from material.
b. Remove all trash and boxes from site.
c. When contracted work is completed, the vendor is responsible for restoring the access and installation areas to their original appearance. Vendor to notify Project Manager of any damages that occur during the work process, to schedule when repairs can be done at the vendor's expense: this includes, but is not limited to, damage or soiling to walls, carpeting, flooring and ceilings.
d. Provide sufficient maintenance and protection to ensure freedom from damage and deterioration at time of substantial completion.
e. Vendor will coordinate with other agencies for smooth execution of the work at site. Also, the vendor to coordinate with Consultants for Clarifications. The single point of contact for the Vendors will be Project Manager.

4.6 HANDING OVER

Vendor to provide the Handing over Documents in Triplicate as per the Formats issued by the Project Manager as per the following List:

a. Completion Certificate (Format to be issued by Project Managers)
b. No Dues Certificate
c. Warranty Certificates
d. Design Concept / Scope of Works
e. Pre-Commissioning / Commissioning / Test Reports
f. Technical Data / Product Catalogues
g. Do & Don'ts List
h. Call Escalation Chart (Contact Details for Emergency)
i. Supplier Contact Details for Miscellaneous Items
j. Maintenance Schedule & Recommended Tools
k. Inventory List / Asset List
l. Recommended Spares List
m. As Built Drawings
n. Copy of WO / PO

The above list is not final and sacrosanct and may be modified as per Client's requirement in keeping with the prevailing Statutory & industry norms. The Vendor accepts that the same shall be provided unconditionally.

4.7 PROJECT DELIVERABLES / TIME LINES

a. Manufacturing Time
b. Shipment time required
c. Port turnaround time (expected)
d. Transportation time required till the site
e. Installation Time on Site

4.8 INSURANCES

Refer Principal Model Standard Terms & Conditions

4.9 COMMENCEMENT TIME AND DELAYS

4.9.1 Commencement of works

The VENDOR shall commence the work on site within **2-3 days** from the date of email confirmation/ Letter of Intent / CBRE's written order to commence the work and shall proceed with the same with due expedition and without delay, except as may be expressly sanctioned or ordered by the Client / CBRE or be wholly beyond the VENDOR's control.

4.9.2 Possession of site

The possession of the Site shall under all circumstances and at all times be with the Client. The BID shall, not under any circumstances be deemed to be giving the VENDOR a lease, license or other possessory right over the Site. The VENDOR and his personnel shall be permitted suitable access to the Site for the purposes of carrying out the works. The Client, may at any time, remove or cause to be removed any of the VENDOR's articles, things, construction plant, personnel or labour without notice to the VENDOR. This shall not, in any manner, prejudice or affect the VENDOR's liabilities and obligations in respect of the works and in particular the liability arising due to any damage to any person or material at the Site.

4.9.3 Way leaves, etc.

The VENDOR shall bear all Prices and charges for special or temporary way leaves required by him in connection with access to the site. The VENDOR shall also provide at his own Price any additional accommodation outside the site required by him for the purpose of the works.

4.9.4 Time for completion

As per tender schedule / program of works approved by the CBRE Project Manager.

a. In the event of VENDOR failing to comply with the overall and individual milestones contained in the time schedule, he shall be liable to pay penalties/Liquidated Damages as per the terms of the Contract.

b. Subject to any requirement in the BID as to completion of any section of the works before completion of the whole, the whole of the works shall be completed, in accordance within the time stated in the BID or such extended time as may be allowed by the Project Manager under his bid.

4.9.5 Extension of time for completion

Refer Principal Model Standard Terms & Conditions

4.9.6 Night or Sunday work

Subject to any provision to the contrary contained in the BID, none of the permanent work shall, save as hereinafter provided, be carried on during the night or on Sundays, if locally recognized as days of rest or their locally recognized equivalent without the permission in writing of the Project Manager or his representative except when the work is unavoidable or absolutely necessary for the saving of life or property or for the safety of the works, in which case the VENDOR shall immediately advise the Project Manager or his representative. Provided always that, the provisions of this clause shall not be applicable; in the event of any work, which is customarily carried out by rotary or double shifts However if the site progress so warrants, the VENDOR shall arrange to carry out night and/ or Sunday works through a dedicated team after obtaining necessary permission for the same.

4.9.7 Rate of progress

Refer Principal Model Standard Terms & Conditions

4.9.8 Delay Damages

Timely completion of the work is the essence of the contract. The VENDOR recognizes and acknowledges that the Client would suffer substantial losses and damage if there were a delay in the execution of the works. If the VENDOR shall fail to achieve completion of the works in progressive stages as mentioned in the conditions of contract, within the permitted time, the VENDOR shall be liable for penalty. However, if the VENDOR compensates it by progressing in other stages without compromising on the quality as adjudged by the project manager, the Client may consider waiver or reduction of penalty in the subsequent stages on specific recommendation by the Project Manager. The penalty for delay shall be 0.2% of the total contract value per day of delay, not limited to project completion but each stage of work, subject to a maximum of 10% of the total contract value.

Notwithstanding the above, if the Client, at any point of time, estimates that a delay in the overall completion of the project, may cause damages to the project, Client's interests / business, the Client upon specific recommendation of the Project Manager, may insist upon a claim against Liquidated Damages as specified in this clause, or at his sole discretion, get the pending works completed by other agencies as appointed by the Client(even if the same is at an escalated price), at the risk and cost of the Vendor. The delays on account of Client, however, shall be excluded from this.

When damages cannot be pre- determined / assessed in advance, then the amount recoverable is said to be 'at large' (to be agreed mutually or determined as specified in clause 21 Settlement of Disputes, in the event of breach).

However, the decision of the Client shall be final and binding on the Vendor, The Client may without prejudice to any other method of recovery, deduct the amount from any monies in his hands, due or which may become due to the VENDOR.

The payment or deduction of such damages shall not relieve the VENDOR from his obligation to complete the works or from any other of his obligations and liabilities under the BID.

4.9.9 Certificate of Substantial Completion of works

When the whole of the works has been completed along with a punch list of items completed and / or corrected, the VENDOR may give a notice of substantial completion to the Project Manager accompanied by an undertaking to finish any outstanding work during the defect's liability period. Such notice and undertaking shall be in writing and shall be deemed to be a request by the VENDOR for the Project Manager to issue a certificate of Substantial Completion in respect of the

works. The Project Manager shall, in consultation with and with prior written consent of the Architects, within 7 days of the date of delivery of such notice either issue to the VENDOR, a certificate of Substantial Completion stating the date on which, in his opinion, the works were satisfactorily completed in accordance with the contract or give instructions in writing to the VENDOR specifying all the work which, in the opinion of the Project Manager, requires to be done by the VENDOR before the issue of such certificate. The Project Manager shall also notify the VENDOR of any defects in the works affecting satisfactory completion that may appear after such instructions and before completion of the works specified therein.

The VENDOR shall be entitled to receive such certificate of Substantial Completion within 15 days of completion to the satisfaction of the Project Manager of the works so specified and making good any defects so notified, provided always that a certificate of Substantial Completion given in respect of any section or part of the permanent works before completion of the whole shall not be deemed to certify completion of any ground or surfaces requiring reinstatement, unless such certificate shall expressly state so.

4.9.10 Provisional acceptance

The work shall deem to have been provisionally accepted after fulfillment of all of the following by the VENDOR: -

- Submitting as-built drawings, O&M Manuals, catalogues, brochures, data-sheets, spares called for in the Contract, etc., and everything else necessary for the proper use and maintenance for the works complete with all systems and services.
- Obtaining certificate of Substantial Completion from the Project Manager
- Handing over of the works to the Project Manager.

4.9.11 Approval and Acceptance

The works shall deem to have been approved and accepted only on issue of Final Completion certificate by the Architect & Project Manager without prejudice to the Client's rights.

4.9.12 Settlement of Disputes

Refer Principal Model Standard Terms & Conditions

FORM OF BID

To,

CBRE South Asia Pvt. Ltd.,
11th & 12th Floor, Raheja Towers, G Block, Bandra Kurla Complex, Bandra (E),
Mumbai 400051, Maharashtra, India

Dear Sir,

Having examined the drawings, general and special conditions of BID, specification and bill of quantities for the execution of the above named works, we, the undersigned, offer to Civil & Infra Works – General Contract Terms in conformity with the said drawings, conditions of BID specification and bill of quantities for the sum of rupees _____ (INR _____) or such other sums as may be ascertained in accordance with the said conditions

We undertake if our Bid is accepted to commence the works immediately from the date of the LOA / email confirmation and to complete and deliver the whole of the works comprised in the contract within **90 calendar days** calculated from the day of LOA.

If our Bid is accepted, we will obtain the guarantee of a scheduled Bank (to be approved by you) for a sum not exceeding 30% (Thirty percent) of the above-named sum for advance of the contract under the terms to be approved by you.

We agree to abide by this Bid for the period of **60 days** from the date fixed for submitting the same and it shall remain binding upon us and may be accepted at any time before the expiration of that period. We would facilitate the timely finalization of the decision by CBRE / Client of the acceptance of the Bid by being present at their Mumbai / Pune offices for meetings as and when requested by the Project Management Consultants.

We confirm that there are no deviations in the Bid documents submitted herewith from the pro-forma Bid documents except for those qualified separately in the offer.

Thanking you,

Yours truly,

Signature of Bidder/ Vendor

Occupational Health, Safety and Environment (OHSE) Requirements

1) INTRODUCTION

This document illustrates the principle requirement of CBRE South Asia Pvt Ltd, on Occupational Health, Safety and environment associated with the Contractor. The CONTRACTOR, his Subs and DC's, shall comply with the safety precautions, protective measures, housekeeping requirements, etc. The Project Managers / Architects with due intimation shall have the right to stop the work at site, if in his opinion proceeding with the work will lead to an unsafe and dangerous condition. The CONTRACTOR shall get the unsafe condition removed and also provide protective equipment. The CONTRACTOR shall ensure that all workmen are aware about the nature of risk involved in their work and have adequate knowledge for carrying out their work safely.

The instructions issued herein are indicative and not exhaustive. Therefore the CONTRACTOR shall be responsible to ensure that adequate safety measures have been adopted in the course of execution of the contract in accordance with safety standards / statutory regulations, as applicable.

The CONTRACTOR shall be held responsible for non-compliance if any of the safety measures and delays, implications, injuries, fatalities and compensation arising out of such situations or incidents.

2) OHSE TARGET AND GOALS

- Zero recordable injuries
- Zero reportable environmental incidents
- 100% adherence to usage of proper PPE at work.
- 100% incident recording and reporting

3) DUTIES/RESPONSIBILITIES OF OHSE COORDINATOR/CONTRACTOR

1) Safety person shall wear green colored helmet while at work.
2) The project Manager shall be responsible for implementing and maintaining safety at work site and the EHS representatives shall advise , assist & provide training and create awareness to all about safe practices at work place Responsibility lies on all team members of the project to achieve the following goals
    a) 100% implementation of safety rules and regulations.
    b) Checking work areas , making housekeeping inspections
    c) Keeping records of all unsafe conditions found and corrective action taken.
    d) Enabling members to use PPE
    e) Conducting safety committee meeting on a periodic basis to discuss on the safety non compliances.

4) SAFETY ORGANISATION/SAFETY COMMITTEE

- A safety committee should be formed at site. The members should be selected from Client, PMC and the Contractor. The Project Manager will be the Chairperson of the committee.
- Meeting should be conducted on a periodic basis with all the safety committee members. MOM should be circulated among the members. Project managers shall ensure the compliance of the issues raised in the meetings as per the target dates mentioned in MOM

5) TRAFFIC

i. The CONTRACTOR shall organize operations by taking approval to use the existing roads, routes / access, if any, from Client.

ii. The CONTRACTOR shall exercise full care to ensure that no damage is caused by him or workmen, during the operation, to the existing water supply, sewerage, power or telecommunication lines or any other services or works. The CONTRACTOR shall provide and erect before construction, substantial barricades, guardrails, and warning signs. He shall furnish, place and maintain adequate warning lights, signals, etc., as required by Project Managers / Architects.

iii. Pedestrian walkways should be segregated from the path of vehicle movement.

iv. Speed limit sign boards should be displayed at site.

v. Speed breakers must be provided so as to reduce the over speed of vehicles.

6) SAFE MEANS OF ACCESS/ EGRESS

i. Adequate and safe means of access and exit shall be provided for all work places, at all elevations.

ii. Suitable scaffolds shall be provided for workmen for all works that cannot safely be done from the ground, or from solid construction except such short duration work as can be done safely from ladders. Ladder shall be of rigid construction having sufficient strength for the intended loads and made of metal and all ladders shall be maintained well for safe working condition. Suitable footholds and handholds shall be provided on the ladder. The ladder shall be given an inclination not steeper than 1 in 4 (1 horizontal and 4 vertical / at an angle of 75 degree to the wall)

iii. Access/Egress should be free of obstructions and from any slippery substances like oil, water or grease.

iv. The landing space for the access/egress should be a leveled surface.

7) SCAFFOLDS

i. Scaffolding or staging more than 3.5m above the ground or floor, swung or suspended from an overhead support or erected with stationary support shall have a standard guard-rail properly attached, bolted, braced or otherwise secured at least 1m high above the floor or platform of such scaffolding or staging. The guardrail shall extend along the entire exposed length of the scaffolding with only such opening as may be necessary for the delivery of materials. Work platforms must have hand rails at 1m height and Mid Rails at 0.5m level. Such scaffolding or staging shall be so fastened as to prevent it from swaying from the building or structure. Scaffolding and ladder shall conform to relevant IS specification (IS 3696-1966). TIMBER/BAMBOO SCAFFOLDING SHALL NOT BE USED.

ii. Working platforms of scaffolds shall have toe boards at least 15cm in ht. to prevent materials from falling down.

iii. A sketch of the scaffolding proposed to be used shall be prepared prior to start of erection of scaffolding. Authorized scaffolder shall examine all scaffolds before using.

iv. Working platform, gangways and stairways shall be so constructed that they shall not sag unduly or unequally and if the height of the platform or gangway or stairway is more than 3.5m above ground level or floor level, they shall be closely boarded, shall have adequate width for easy movement of persons and materials and shall be suitably guarded.

v. The planks/jalley used for working platform shall not project beyond the end supports to a distance exceeding four times the thickness of the planks used. The planks/jalleys shall be rigidly

tied at both ends to prevent sliding and slippage. The thickness of the planks/jalleys shall be adequate to take load of men and materials and shall not collapse.

vi. Every opening in the floor of a building or in a working platform shall be provided with suitable means to prevent fall of persons or materials by providing suitable barricades, fencing or railing, the minimum height of which shall be 1m, along with 15cm high sheet obstruction at floor level along the railing.

vii. Safe means of access shall be provided to all working platforms and other elevated working places. Every ladder shall be securely fixed. No single portable ladder shall be over 9m in length. For ladders, up to 3m in length the width between side rails in the ladders shall in no case be less than 300mm. For longer ladders, this width shall be increased by at least 20mm for each additional meter of length. Step spacing shall be uniform and shall not exceed 300mm.

viii. Adequate precautions shall be taken to prevent danger from electrical lines and equipment. No scaffolding, ladder, working platform, gangway runs, etc. shall exist within 3 meters of any un-insulated electric wire. Whenever electric power and lighting cables are required to run through (pass on) the scaffolding or electrical equipment are used, such scaffolding structures shall have minimum two earth connections with earth continuity conforming to IS code of practice.

ix. Bottom of the scaffold should be provided with base plates. Proper cross bracings should be given for supporting the scaffold.

8) DEMOLITION

i. Before any demolition work is commenced and also during the progress of the work:

ii. All roads and open area adjacent to the work site shall either be closed or suitably protected. Appropriate warning signs shall be displayed for cautioning approaching persons.

iii. Before demolition operations begin, the CONTRACTOR shall ensure that job safety analysis is prepared and take approval from CBRE/Client & the power on all electric service lines is shut off and the lines cut or disconnected at or outside the demolition site. If it is necessary to maintain electric power during demolition operation, the required service lines shall be adequately protected against damage with a licensed electrician only All Persons including those handling heavy materials/equipment shall also wear all basic PPEs at the time of entire operations

iv. Contractor shall submit waste management plan and only after approval by client the demolishing activity will be carried out. No floor, roof or other part of the building shall be overloaded with debris or materials as to render it unsafe. In case of any noncompliance of the waste management plan contractor will be liable for disciplinary procedures

v. Entries to the demolition area shall be restricted to authorized persons only.

vi. Adequate caution signage's, effective gate management shall be ensure by the contractor while carrying out the job.

9) PERSONAL PROTECTIVE EQUIPMENT

i. All necessary personal protective equipment shall be kept available for the use of the persons employed on the site and maintained in a condition suitable for immediate use. Also the CONTRACTOR shall take adequate steps to ensure proper use of equipment by those concerned. The personal protective equipment to be provided by the CONTRACTOR is detailed below.

ii. All persons employed at the construction site shall use safety helmets. No worker will be allowed entry to the site without safety helmets and ID cards.

iii. Workers employed on mixing asphaltic materials, cements and lime mortars shall use protective goggles, protective footwear and hand gloves.

iv. Persons engaged in welding and gas-cutting works shall use suitable aprons, leather gloves and welding face shields. The persons who assist the welders shall use suitable goggles. Protective goggles shall be worn while chipping and grinding.

v. Stonebreakers shall use protective Goggle, Face shield & Nose Mask. They shall be seated at sufficiently safe intervals of distance.

vi. Persons engaged in or assisting in shot blasting operations and cleaning the blasting chamber shall use suitable gauntlets, overalls, dust-proof goggles, boots and protective hood supplied with fresh air at the minimum rate of 9m/hr.

vii. All persons working at height more than 2.5m above ground or floor and exposed to risk of falling down shall use full body safety harness, unless otherwise protected by cages, guard railings, etc. In places where the use of safety belts is impractical, suitable net of adequate strength fastened to substantial supports shall be employed.

viii. Protective nets shall be provided at places where men are working at heights to safe guard against fall of men or material including debris. Contractor shall provide head protection where height works are carried out where workers movement cannot be restricted or re directed.

ix. When workers are employed in sewers and inside manholes that are in use, the CONTRACTOR shall ensure that the manholes are opened and are adequately ventilated at least for an hour. After it has been well ventilated, the atmosphere inside the space shall be checked for the presence of any toxic gas or oxygen deficiency and recorded in the register before the workers are allowed to get into the manholes. The manholes opened shall be cordoned off with suitable railing and provided with warning signals or caution boards to prevent accidents. Contractor shall submit the method statement and job safety analysis and the work can be allowed to carry out only after approval of CBRE/Client. Confined space activities shall be carried out by the contractor after obtaining work permit from the client. Night work in such areas will be allowed only if all the contents of the work permit are compiled and close monitoring is ensured.

x. In case the CONTRACTOR fails to supply the safety equipments, the owner may purchase the same and debit the Price of the equipments and incidentals, to the CONTRACTOR.

S #	Name of Safety Equipment/ Accessories
1	Safety Helmet (Design for carrying load)
2	Regular Safety Helmet / Hard Hat
3	Industrial Safety Shoes
4	Safety Gum Boot
5	Safety Goggles
6	Safety Gloves (Cotton)
7	Safety Gloves (Electrical)
8	Safety Gloves (Leather)

9	Safety Gloves (Rubber)
10	Particulate Respirator
11	Ear plug
12	Full Body Harness
13	Safety Nets -12mm; 10m x 3m
14	Safety Rope
15	Single Sided Barricading Tape Red & White / Yellow & Black
16	Laminated Safety Poster 18" x 12" [1 For 100 Sq. M]
17	Fire Extingisher-CO2-4.5 Kg
18	Fire Extingisher-ABC-5Kg
19	Sand Bucket
20	Fire Blanket (1mtrX1mtr)
21	Aluminum Ladder

10) PAINTING

i. The CONTRACTOR shall not employ women on the work of painting with products containing lead in any form. Only men above the age of 18 years shall be employed on the work with lead paint. The following precautions shall be taken during the work:

ii. Supply air respirators for use by the workers when paint is applied in the form of spray, or a surface having lead paint is scraped.

iii. The workmen shall use overalls and adequate facilities shall be provided to enable the painters to wash at the cessation of work.

iv. All painting jobs, especially those in which lead paints are used shall be kept under industrial hygiene surveillance.

v. Smoking, open flames or sources of ignition shall not be allowed in places where paints and other flammable substances are stored, mixed or used. A caution board, with the instructions written in national/regional language, "SMOKING - STRICTLY PROHIBITED" shall be displayed in the vicinity where painting is in progress or where paints are stored. Symbols shall also be used for caution boards.

vi. Suitable fire extinguishers/sand buckets shall be kept available at places where flammable paints are stored, handled or used.

vii. When painting work is done in a closed room or in a confined space, adequate ventilation shall be provided. If adequate ventilation cannot be provided, workers shall wear suitable respirators.

viii. Epoxy resins and their formulations used for painting shall not be allowed to come in contact with the skin. The workers shall use protective gloves or suitable barrier creams.

ix. Adequate ventilation shall be provided especially when working with hot resin mixes.

x. Increased personal hygiene shall be practiced to control inadvertent contact with the resin and eliminate its effects.

xi. Workers shall thoroughly wash hands and feet before leaving the work. Work clothes shall be changed and laundered frequently.

11) LIFTING MACHINES AND TACKLES

i. Use of lifting machines and tackles including their attachments, anchorage and supports shall confirm to the following standards or conditions:

ii. Lifting machines and tackles shall be of good mechanical construction, sound material and adequate strength and free from any defects and shall be kept in good repair and in good working order.

iii. Every rope used in hoisting or lowering materials or as a means of suspension shall be of good quality and adequate strength and free from any defect.

iv. Every crane operator or lifting appliance operator shall be properly qualified. No person under the age of 21 years shall be in charge of any hoisting machine or give signal to operator of such machine.

v. In case of every lifting machine (and of every chain, ring, hook, shackle, swivel and pulley block used in hoisting or as means of suspension) the safe working load shall be ascertained and clearly marked. In case of a lifting machine having a variable safe working load, each safe working load and the conditions under which it is applicable shall be clearly indicated. No part of any machine or any gear referred to above in this paragraph shall be loaded beyond the safe working load except for the purpose of testing. The safety engineer shall approve this.

vi. The safety engineer shall note the safe working load. Regarding other machines, the CONTRACTOR shall notify the safe working load of the machine to the safety engineer, whenever he brings any machinery to site of work and gets it verified by the safety engineer.

vii. Thorough inspection and load testing of lifting machines and tackles shall be done by a competent person at least once every 6 months and records of such inspection and testing shall be maintained.

viii. Motors, gearing transmission, couplings, belts, and chain drives and other moving parts of hoisting appliances shall be provided with adequate safeguards. Hoisting appliances shall be provided with such means as will reduce to the minimum the risk of any part of a suspended load becoming accidentally displaced or lowered. Barricades shall be erected around the place of hoisting the equipment(s).

ix. Load shall be never allowed to swing during lifting.

x. Tires should be inspected for cuts, tears, breaks and proper inflation.

xi. Copy of third party test certificate for slings should be submitted to the safety dept.

12) WELDING AND GAS CUTTING

i. Welding and gas cutting operations shall be done by qualified and authorized persons and as per IS specifications and code of practice.

ii. Welding and gas cutting shall not be carried out in places where flammable or combustible materials are kept and where there is danger of explosion due to presence of gaseous mixtures.

iii. Welding and gas cutting equipment including hoses and cables shall be maintained in good condition.

iv. Barriers shall be erected to protect other persons from harmful rays from the work. When welding or gas cutting is done in elevated positions, precautions shall be taken to prevent sparks or hot metal falling on persons or flammable materials.

v. Suitable type of protective clothing consisting of fire resistant gauntlet gloves, leggings, boots and aprons shall be provided to workers as protection from heat and hot metal splashes. Welding shields with filter glasses of appropriate shade shall be worn as face protection.

vi. Adequate ventilation shall be provided while welding in confined space or while brazing, cutting or welding zinc, brass, and bronze, galvanized or lead coated materials.

vii. Welding and gas cutting shall not be done on drums, barrels, tanks or other containers unless they have been emptied, cleaned thoroughly and it is made certain that no flammable material is present.

viii. Fire extinguisher shall be available near the location of welding operations. Fire safety permit shall be obtained for working at vulnerable areas and operating areas before flame cutting/welding is taken up.

ix. For electric (Arc) welding the following additional safety precautions shall be taken:

x. When electrical welding is undertaken near pipelines carrying flammables, such pipelines shall not be used as part of earth conductor, but a separate earth conductor shall be connected to the machine directly from the job.

xi. Personnel contact with the electrode or other live parts of electric welding equipment shall be avoided.

xii. Extreme caution shall be exercised to prevent accidental contact of electrodes with ground.

xiii. The welding cable shall not be allowed to get entangled with power cables. It shall be ensured that movement of materials does not damage the cables.

xiv. For gas cutting the following safety precaution should be taken:

a) Four numbers of Flash back arrestors should be present as two for cylinders and two for torch

b) The sign of flashback will be a loud crack, followed by a puff of carbon from the welding tip when relighting.

c) The consequence of not arresting a flashback can range from bursting of hose to a bombing explosion of cylinder.

d) The least damage a flashback can do is to deposit carbon in torch valves which can affect their operation and can lead to more serious flashbacks.

e) Never join acetylene hoses with copper tube or copper wire since they may produce an explosive mixture.

f) Never set the acetylene cylinder outlet gauge pressure more than 100kPa

g) Acetylene smells like onion or garlic, which helps in detecting leak.

13) GRINDING

i. All portable grinders shall be used only with their wheel guards in position to reduce the danger from flying fragments should the wheel break during the use.

ii. Grinding wheels of specified diameter only shall be used on a grinder – portable or pedestal - in order not to exceed the prescribed peripheral speed.

iii. Goggles shall be used during grinding operation.

iv. Guard in the wheel shall never be removed during work.

v. Grinding the material on the sides of the wheel can damage the wheel.

14) DRILLS:

a. Care should be taken not to drill too large hole at a time.

b. Drilling of large holes should be done in stages by using small bore drills and gradually increasing the size as necessary.

c. Bit should be properly tightened to the chuck.

d. Auxiliary handle should be used for continuous work.

e. Do not drill with one hand while holding the material with other.

f. Do not raise or lower the drill by its power cord.

15) HOUSE KEEPING

i. The CONTRACTOR shall at all-time keep his work site, site office and surroundings clean and tidy from rubbish, scrap, surplus materials and unwanted tools and equipment.

ii. Welding and other electrical cables shall be so routed as to allow safe traffic by all concerned.

iii. No materials on any of the sites of work shall be so stacked or placed as to cause danger or inconvenience to any person or the public.

iv. At the completion of the work, the CONTRACTOR shall ensure removal from the work premises all scaffoldings, surplus materials, rubbish and all huts and sanitary arrangements used/installed for workmen on the site.

v. Any rubbish materials should not be burnt at site.

16) ALCOHOL and DRUGS

1) The contractor shall ensure at all times that no Labour is working under the influence of alcohol/drugs. If found they should be suspended from the site immediately.

2) Smoking at worksite by any member is also prohibited.

17) FIRE SAFETY

i. All necessary precautions shall be taken to prevent outbreak of fires at the construction site. Adequate provisions shall be made to extinguish fires, should they still break out.

ii. Quantities of combustible materials like timber, bamboo, coal, paints, etc. shall be the minimum required in order to avoid unnecessary accumulation of combustibles at site.

iii. Containers of paints, thinners and allied materials shall be stored in a separate room, which shall be well ventilated, and free from excessive heat, sparks, flame or direct rays of the sun. The containers of paint shall be kept covered and properly fitted with lid and shall not be kept open except while using.

iv. Fire extinguishers shall be located at the construction site at appropriate places.

v. Adequate number of workmen shall be given education and training in firefighting and extinguishing methods.

18) WORK IN RADIATION AREA

The CONTRACTOR shall follow the stipulated procedure regarding work in the radiation are and other works related with radiography.

19) WORK IN AND AROUND WATER BODIES

When the work is done near any places where there is risk of drowning, all necessary rescue equipment such a life buoys and life jackets shall be provided and kept ready for use and all necessary steps taken for prompt rescue of any person in danger and adequate provision shall be made for prompt first-aid treatment of all injuries likely to be sustained during the course of the work. Persons who do not know swimming shall not be engaged alone for any work where risk of drowning exists. Sufficient number of life buoys or life jackets shall be provided.

20) MEDICAL FACILITIES

i. The CONTRACTOR shall arrange for medical aid and treatment for his staff and workers engaged on the work site including the first-aid facilities if they are not available at the project site.

ii. First-aid appliance including sterilized dressing, cotton wool and antiseptic cream shall be made available at readily accessible places at every work site. These shall be maintained in good order under the charge of a responsible person.

iii. Each first aid box shall contain the following equipment and adequate to the number of workers

(As per The BOCW Act 1996)

1) A sufficient number of eye wash bottles filled with distilled water or suitable liquid clearly Indicated by a distinctive sign which shall be visible at all times.
2). 4% xylocaine eye drops, and boric acid eye drops and soda bicarbonate eye drops.
3). 24 – Small sterilized dressings.
4). 12 – Medium size sterilized dressings.
5). 12 – Large size sterilized dressings.
6). 12 – Large size sterilized burn dressings.
7). 12 – (Fifteen cms) packets of sterilized cotton wool.
8). (200 ml) bottle of centrimide solution (1%) or suitable antiseptic solution.
9). One (200 ml) bottle of Mercurochrome solution (2%) solution in water.
10). One (120 ml) bottle of salvolatile having the doses and mode of administration indicated on the label.
11). One pair of Scissors.
12). One roll of adhesive plaster (6 cm x 1metre).
13). Two rolls of adhesive plaster (2 cm x1 meter).
14). 12 pieces of sterilized eye pads in separate sealed packets.
15). A bottle containing 100 tablets (each of 325 mg) of aspirin or any other analgesic.
16). 12 – roller bandages 10 cms wide.
17). 12 – Roller bandages of 5 cms wide.
18). One tourniquet.
19). A supply of suitable splints.
20). 3 packets of safety pins.

21). Kidney tray.
22). A snake bite lancet.
23). One (30 ml) bottle containing potassium permanganate crystals.
24). Copy of First-aid leaflet issued by Director General.
25). 6 – Triangular bandages.
26). 2 pairs of suitable, sterilized, latex hand gloves.

iv. At large work places where hospital facilities are not available within easy reach of the work, first-aid posts shall be established. Ambulance availability shall be identified during the entire period of work for attending to injury cases.

21) SAFETY COORDINATOR

The CONTRACTOR shall have a Safety Officer or a Supervisor to be designated as a Safety Coordinator in order to specifically look into the implementation of various safety requirements of the site work. The person thus designated will in general co-ordinate on matters of safety and in particular ensure that the Safety Manual is complied with. His name shall be displayed on the Notice Board at a prominent place at the work site.

22) LABOUR WELFARE

1) The contractor shall provide separate urinals/toilets for male and female workers as per the guidelines given in BOCW Act-1996
2) Drinking water should be made available for the workers at site The drinking water should be tested for every six months and the report should be submitted by the contractor to CBRE
3) In every workplace wherein not less than 100 workers are ordinarily employed, the contractor shall provide an adequate canteen for the workers.
4) The contractor shall provide temporary accommodation to all workers. These accommodations shall have bathing and washing & good sanitation facilities including rest rooms.

23) REPORTING OF ACCIDENT

i. All accident leading to property damage and/or personnel injuries shall be reported to the concerned authorities' viz. Insurance Co. Police, Head Office, Regional Office, etc.

ii. The CONTRACTOR shall also submit a monthly statement of accidents to the Project Managers / Architects by every month showing details of accident, nature of injury including disability, days lost, treatment provided, etc., and the extent of property damage.

iii. The statutory safety records required to be maintained by the CONTRACTOR at the site are
   a. Accident Register
   b. Dangerous Hazards Register
   c. First Aid Register
   d. Register for the inspection notes by government officers.
   e. Copies of all accident reports and hazard reports filed to the prescribed authorities.
   f. Incident/Near-miss registers

24) PUBLIC PROTECTION

The CONTRACTOR shall make all necessary provisions to protect the public. He shall be held responsible for defense of every action of other proceedings at law that may be brought by any person for injury sustained owing to neglect of any precaution required to taken to protect the public.

25) OTHER STATUTORY PROVISIONS

All operations involving the transport, handling, storage and use of explosive shall be as per the standing instructions and conform with the latest Indian Explosives Act and the explosives Rules. Handling, transport, storage and use of compressed gas cylinders and pressure vessels shall conform to the latest Gas Cylinder Rules and Static and Mobile Pressure Vessels (Unfired) Rules. In addition, The Indian Electricity Act and Indian Electricity Rules - latest, the Atomic Energy Act, the Radiation Protection Rules - latest, Radiation Protection Manual of Nuclear Facilities and the Atomic Energy (Factories) Rules – latest, and various latest rules and Act related to mining shall also be strictly complied with.

26) GUIDELINES AND GENERAL PROCEDURES FOR SUPPLY AND USE OF ELECTRICITY AT SITE

1. **GENERAL**

Following safety requirements shall be complied with before the CONTRACTOR uses the power supply.

1. The CONTRACTOR shall submit a list of licensed electrical staff to be posted at Site.
2. It shall be the responsibility of the CONTRACTOR to provide and maintain complete installation on the load side of the supply point with regard to the safety requirements at Site. All cabling and installation shall comply with the appropriate latest statutory requirements given below and shall be subject to approval of the Engineer-in-Charge:
    a. Indian Electricity Act.
    b. Electricity (Supply) Act.
    c. Indian Electricity Rules.
    d. National Electricity Code.
    e. Other relevant rules of Local Bodies and Electricity Boards.
3. The power supply shall be regulated as per the terms and conditions of the supply of the respective electricity boards.
4. Where distribution boards are located at different places the CONTRACTOR shall submit schematic drawing indicating all details like size of wires, Overhead and Cable feeders, earthing etc. The position and location of all equipment and switches shall be given.
5. The CONTRACTOR shall make his own arrangement for main earth electrode and tapings thereof. The existing earth points available at site can be used at the discretion of Client with prior permission. Method of earthing, installation and earth testing results shall conform to relevant I.S. Specifications (IS-3043).
6. All three phase equipments shall be provided with double earthing. All light fixtures and portable equipment shall be effectively earthed to main earthing.
7. All earth terminals shall be visible. No gas pipes and water pipes shall be used for earth connection. Neutral conductor shall not be treated as earth wire.
8. The CONTRACTOR shall not connect any additional load without prior permission of Client.

9. Joints in earthing conductors shall be avoided. Loop earthing of equipment shall not be allowed. However, tapings from an earth bus may be done.

10. The entire installation shall be subjected to the following tests before energizing of installation including portable equipment: -

    a. Insulation resistance test.

    b. Polarity test of switches.

    c. Earth continuity test.

    d. Earth electrode resistance.

    The test procedures and their results shall conform to relevant standards.

2. **INSTALLATION**

1. Only persons having valid wireman's license/competency certificate shall be employed for carrying out electrical work and repair of electrical equipment, installation and maintenance at site. A qualified licensed Supervisor shall supervise the job.

2. Electrical equipment and installations shall be installed and maintained as to prevent danger from contact with live conductors and to prevent fires originating from electrical causes like short circuits, overheating etc. Installation shall not cause any hindrance to movement of men and materials.

3. Materials for all electrical equipment shall be selected with regard to working voltage, load and working environment. Such equipment shall conform to the relevant standards.

4. The minimum clearance to be maintained for all overhead lines along roads and across roads shall be as per the statutory requirements.

5. Grounding conductor of wiring system shall be of copper or other corrosion-resistant material. An extra grounding connection shall be made in appliances/equipment where chances of electric shock are high.

6. Electric fuses and/or circuit breakers installed in equipment circuits for short circuit protection shall be of proper rating. It is also recommended that high rupturing capacity (HRC) fuses are used in all circuits. Earth Leakage Circuit Breaker (ELCB) /Residual Current Circuit Breaker (RCCB) of 30 mA sensitivity shall be provided in the circuits.

7. Wherever cables or wires are laid on poles, a guard wire of adequate size shall be run along the cables/wires and earthen effectively. Metallic poles as a general rule, shall be avoided and if used shall be earthen individually. Anti-climbing guards and danger notices shall be provided on poles. Each equipment shall be an individual isolating switch.

8. Wires and cables shall be properly supported and an approved method of fixing shall be adopted. Loose hanging of wires and cables shall be avoided. Lighting and power circuits shall be kept distinct and separate.

9. Reinforcement rods or any metallic part of structure shall not be used for supporting wires and cables, fixtures, equipment, earthing etc.

10. All cables and wires shall be adequately protected mechanically against damages. In case the cable is required to be laid underground, it shall be adequately protected by covering the same

with bricks, Plain Cement Concrete (PCC) tile or any other approved means and Cable marker shall be displayed as per local standards.

11. Using suitable cable glands shall properly terminate all armoured cables. Using cable lugs/sockets shall connect multi-stranded conductor cables. Cable lugs shall preferably be crimped. They shall be of proper size and shall correspond to the current rating and size of the cable. Twisted connections will not be allowed.

12. All cable glands, armoring and sheathing of electric cable, metal circuits and their fittings, metallic fittings and other non-current carrying parts of electrical equipment and apparatus shall be effectively grounded.

13. All the Distribution Boards, Switch Fuse Units, Bus bar chambers, ducts, cubicles etc. shall have MS enclosures and shall be dust, vermin and waterproof. The Distribution Boards, switches etc. shall be so fixed that they shall be easily accessible. Changes shall be done only after the approval of the Engineer-in-Charge.

14. The CONTRACTOR shall provide weather proof metal enclosures/covers for protection of the entire switchboard, equipment. Exposed live parts of all electrical circuits and equipment shall be enclosed permanently. Crane trolley wires and other conductor which cannot be completely insulated shall be placed such that they are inaccessible under normal working conditions.

15. Iron-clad industrial type plug & sockets with Cap are preferred for additional safety.

16. PDBs shall have emergency contact details, Signage, incomer details.

17. Isolating switches shall be provided close to equipment for easy disconnection of electrical equipment or conductors from the source of supply when repair or maintenance work has to be done on them.

18. In front of distribution boards a clear space of 90 cm shall be maintained in order to have easy access during an emergency.

19. Adequate working space shall be provided around electrical equipment, which requires adjustment or examination during operation.

20. As far as possible electrical switches shall be excluded from a place where there is danger of explosion. All electrical equipment such as motors, switches and lighting fittings installed in workroom where there is possibility of explosion hazard shall be explosion proof.

21. All connections to lighting fixtures, starters or other power supplies shall be provided with PVC insulate, PVC sheathed twin/three/four core wires to have better mechanical protection for preventing possible damage to equipment or injury to personnel. Taped joints shall not be allowed and the connections may be made in looping system. Electric starter of motors, Switches shall not be mounted on wooden boards. Only sheet steel mounting or iron framework shall be used.

22. All the lighting fixtures and lap holders shall be of good quality and in good condition. Badly repaired or broken holders, etc. shall not be used.

23. Only PVC insulated and PVC sheathed wires or armored PVC insulated and sheathed cables shall be used for external power supply connections of temporary nature. Weatherproof rubber wires shall not be used for any temporary power supply connections. Taped joints in the wires shall not be used.

24. The bulbs/lamps used for illumination and testing purpose shall have cover or guard to protect them from accidental breakage. Only 24 V supply system shall be used for hand lamps etc. while working inside metallic tanks or conducting vessels.

25. Power cable Joints shall be avoided; Unavoidable Joints shall be connected through Cable connector

3. OPERATION & MAINTENANCE

1. All persons, who work with electrical installation/equipment, shall be aware of the electrical hazards, use to protective devices and safe operational procedures. They shall be given training in firefighting, first aid and artificial resuscitation techniques.

2. The CONTRACTOR shall instruct the workers in the proper procedure, specify and enforce the use of necessary protective equipment such as adequately insulated pliers, screw drivers, fuse pulleys, testing lamps and similar hand tools. Only wooden ladders shall be used to reach the heights in electrical work.

3. No material or earthwork shall be allowed to be dumped below or in the vicinity of the bare overhead line conductors.

4. Before any maintenance work is commenced on electrical installations/equipment, the circuits shall be de-energized and ascertained to be dead by positive test with an approved voltage-testing device. Switches shall be tagged or the fuse holders withdrawn before starting the work. Adequate precautions shall be taken in two important aspects viz.

    a. That there shall be no danger from any adjacent live parts and

    b. That there shall be no chances of re-energizing of the equipment on which the persons are working.

5. While working on or near a circuit, whenever possible the use of one hand may be practiced even though the circuit is supposed to be dead. The other hand may preferably be kept in pocket.

6. When it is necessary to touch electrical equipment (for example when checking for overload of motors) back of the hand may be used. Thus, if accidental shock were to cause muscular contractions, one would not 'freeze' to the conductor.

7. Operations of electrical equipment shall be avoided which standing on wet floor or when hands are wet.

8. Before blown fuses are replaced, the circuit shall be locked out and an investigation shall be made for the cause of the short circuit or overload.

9. When two persons are working within reach of each other, they shall never work on difference phases of the supply.

10. When structural repairs, modification or painting work are to be undertaken, appropriate measures shall be taken for the protection of persons whose work may bring them into the priority of live equipment/circuit.

11. It shall be ensured that the insulation and wire size of extension cords are adequate for the voltage and current to be carried.

12. While tapping electricity from the socket, plug top must be used. It shall be ensured that no extension boards are over loaded while tapping. Only standard three pin plugs shall be used for tapping electricity. Broken sockets/plugs shall be replaced immediately with good ones. Only joints free cables shall be used for connecting equipment/ apparatus.
13. Floors shall be kept free from tailing electrical cables to avoid tripping hazard.
14. Power supply to the entire machines and lighting fixture shall be switched off when not in use.
15. Temporary electrical connections shall be removed as soon as the stipulated work is over. After completion of the works, the CONTRACTOR shall dismantle the distribution boards and the other facilities erected at site.
16. Unauthorized/excess tapping of power from distribution boards under the control of the main contractor or client without prior permission is prohibited at all circumstances.
17. No flammable materials shall be stored in any working area near & near the switchboards where there is a possibility of spark or fire
18. "MEN ON LINE" "DO NOT SWITCH ON" "DANGER" OR "CAUTION" boards as applicable shall be used during maintenance works on the electrical equipment.

4. **PORTABLE ELECTRICAL EQUIPMENT**

1. Portable electrical equipment shall be regularly examined, tested and maintained to ensure that the equipment and its leads are in good order. Register shall be maintained for inspection, recording the testing dates and results of the equipment.
2. All portable appliances (with metal body) shall be provided with three-core cable and three-pin plug. The third pin of the plug shall invariably be earthen. It shall be ensured that the metal part of the equipment shall be effectively earthen.
3. All connections to portable equipment or machines from the panel/distribution board/extension board shall be taken using 3 core double insulated PVC flexible copper wire in one length. No joints shall be allowed in this flexible wire. In case single length of wire is not sufficient for a particular location then the supply can be tapped by providing another extension board or using industrial cable connectors comprising of switch and socket.
4. Flexible cables for portable lamps, tools and apparatus shall be regularly examined, tested periodically and maintained to ensure safety.

27) SAFETY CODES IN CONSTRUCTION INDUSTRY

27.1. EXCAVATION (IS-3764)

i. Location and protection of underground utility–viz. Water main, cables.
ii. Barricade – openings.
iii. Lamp at night – warning signs.
iv. Excavated material – 5 ft. Away
v. Shoring for sides – avoid sides from collapsing.
vi. Avoid vertical cuts.

vii. Water controlled.
viii. Equipment / vehicles at safe distance.
ix. Ensure stability of other structures nearby.
x. Do not take gas cylinders inside trenches.
xi. Trenches greater than 4 ft. Depth should be provided with shoring – ladder for each 15m for hazardous area and each 30m for low hazardous area.
xii. Excavated portion.

27.2. CEMENT WORKS

Cement dermatitis, burns-use gumboots, gloves, and respirators.

27.3. SCAFFOLDING (IS 4014(PART II))

i. Clear the area of unwanted materials.
ii. Erection under proper supervision.
iii. Ensure ropes & cables are in good condition.
iv. Ensure that all structural members and all connections adequate.
v. Supports strong – adequate cross bracing.
vi. Ensure ground is safe and provide proper foot hold.
vii. Keep ladder/working area free of grease/oil.
viii. Provide guardrails & toe board.
ix. Wear safety belt, helmet while working on scaffolding.
x. Do not use the scaffolding for more than 15 days without rechecking.
xi. Frequent inspection.
xii. Excavation is not permitted near base of scaffolding.
xiii. Use of wooden scaffolding is not permitted

27.4. LADDERS (IS 3696 (PART II)

i. Proper inspection – rungs/steps are not spliced.
ii. Properly secured – top & bottom.
iii. Side rails on fixed ladders to extend above top landing.
iv. Build up ladders of sound material.
v. Rungs not to exceed 12 inches.
vi. Step ladders fully open during use.
vii. Metal ladders – prohibited near electrical lines. Proper maintenance and storage after use.
viii. Rails of the ladder placed at the roof should extend minimum 1.05m (Vertical height from work Platform to Top of the Ladder).

27.5. BARRICADES

i. Floor openings covered/barricaded properly.
ii. Road ways/sidewalks protected.
iii. Traffic controlled.
iv. walkways
v. Excavation of trenches
vi. Hoisting/Lifting scene
vii. Hazardous areas like radiation test area, Demolishing area, Deshuttering area, Electrical hazardous area, Painting Yard

27.6. CONCRETE CONSTRUCTION

i. Forms properly installed and braced.
ii. Proper curing period & procedure.
iii. Mixing & transporting equipment properly supported/routed.
iv. Adequate runways.
v. Protection from cement dust.
vi. Hard hats – safety shoes, skin covering.
vii. Nails to be removed from shuttering material.

27.7. MASONRY

i. Proper scaffolding.
ii. Dust protection.
iii. Safe hoisting equipment.

27.8. HOISTS, CRANES, DERRICKS

i. Inspect – cables, slings, chains, hooks, eyes
ii. Equipment stability/supports.
iii. Out riggers used if required.
iv. Power lines removed/inactivated (cranes).
v. Signals understood and observed.
vi. Experienced operators; medical fitness , color vision test certificate for operators & signal men
vii. All equipment properly lubricated/maintained.
viii. Protective head gears.
ix. If person climbs on derrick installed for lifting material, special precautions should be taken.
x. Periodical test/inspection of hoists and cranes.
xi. Mark “safe working loads” on hoists/cranes.
xii. Cranes to be operated by certified operator.
xiii. Ensure that slings are vertical.
xiv. Do not drag chains, slings, hooks and load over the floor.
xv. Know the load before lifting.
xvi. Barricade around the area to avoid labour movement close to the area
xvii. Guide ropes are mandatory while lifting materials with different cranes

27.9. CONVEYORS:

i. Proper inspection & maintenance.
ii. Screens and other protection.
iii. Adequate inspection and maintenance of ladders, light
iv. Emergency Stop Pull Cord

27.10. HEAVY EQUIPMENT & TRUCKS

i. Qualified operators with license.
ii. Vehicle laws & regulations to be observed.
iii. Check – breaks, lights, warning devices and wheels.
iv. Weight limits – load sizes controlled.
v. Haul loads well maintained – properly laid.
vi. Personnel not carried in unsafe manner.
vii. Protection when equipment not used.
viii. Planned inspection – maintenance.
ix. Adequate equipment records

x. Proper oil, fuel, lubricants used.
xi. Reverse horn to be fixed for vehicles being used in the site.
xii. Speed limit signage to be installed.
xiii. Fitness certificate of the vehicles

27.11. HOUSE KEEPING & SANITATION

i. Proper and organized material stacking
ii. General neatness in working area.
iii. Regular disposal of waste/trash.
iv. Pathways and walkways clear.
v. Adequate lighting.
vi. Sanitary facilities – clean.
vii. Adequate drinking water.
viii. Scrap yard to be marked

27.12. EMERGENCY PROCEDURE

i. First aid station – properly manned.
ii. First aid boxes – with First aid materials
iii. Nearest Hospital tie up for emergency treatment
iv. All Injuries reported promptly.
v. Adequate emergency evacuation facilities.
vi. Emergency assembly point /Signage / illuminated access
vii. Emergency evacuation procedure & Emergency Response Plan
viii. Training for emergency response teams periodically
ix. Emergency siren , Mega Phone , Whistle , Hand Baton , Reflective Jacket

27.13. WELDING – CUTTING

i. Qualified & experienced technicians
ii. Proper screen, shield, goggles, gloves, clothing, equipment.
iii. Electrical equipment – grounded – inspected.
iv. Power cables protected & in good condition.
v. Fire extinguishers – available.
vi. Inspection of fire hazards.
vii. Flammable materials – protected (40 feet away).
viii. Gas cylinders – upright & chained.
ix. Gas line, torch in good condition.
x. Trolleys at the time of moving cylinders in the working areas
xi. Cylinder stand for storage in Cylinder Yard
xii. Proper covering of mezzanine holes or barricades for cut portion.
xiii. Barricades/notices below mezzanine being cut – work permit
xiv. Helper also to use goggles.
xv. Flash Back arresters to be provided at both end of cylinders and torch

27.14. FLAMMABLE GASES – LIQUIDS

i. "No smoking" signboard.
ii. Containers clearly identified/marked.
iii. Proper storage practices.

iv. Proper storage temperature – protection.
v. Fire hazards to be checked.
vi. Proper & adequate fire extinguishers.
vii. Neat storage area – clear passages.
viii. Material firmly stacked – not too high.
ix. Entry restricted.
x. Store in separate enclosed area.

27.15. HANDLING AND STORAGE

i. Proper number for operation.
ii. Person picking up leads correctly.
iii. Materials protected from heat/moisture.
iv. Protection from falling into hoppers/bins.
v. Dust protection observed.
vi. Extinguishers/fire protection available.
vii. Traffic routing and control.
viii. Safety precautions in line with MSDS- Material safety data sheet

27.16. POWER TOOLS

i. Good housekeeping where used.
ii. Tools – cords – earthing – in good condition.
iii. Proper instructions for use.
iv. Proper mechanical safe guards.
v. Tools nearly stored when not in use.
vi. Right tool for job.
vii. Proper wiring.
viii. Proper training to operator.
ix. Proper supervision.
x. Use of safety appliances - goggles - face shield.
xi. Flying hazards checked up.
xii. Work shall be carried out after obtaining work permit

27.17. EDUCATING LABOUR:

i. Safety Posters/boards to be installed at site at different locations in local languages
ii. Safety Officer to conduct safety meets periodically and holds pep talks with workmen.
iii. Mock drills & fire drills
iv. Awards for motivating Labours
v. Tool box talks,
vi. specialized training for high risk activities

28) SCHEDULE OF FINES

Schedule of Charges to Contractors who are in breach of the employer's Site Safety, Site Safety Cycle and Environmental Rules and Regulations

Noncompliance of content of this document content may result in penalty for first violation, stoppage of work, removal of supervisor or member from the work premises for second violation.

1	Smoking in an unauthorized area and/or consumption of alcohol and/or use of illegal substances.	Rs 500.00
2	Burning of waste or smoldering of combustible materials on site other than for heat treatment processes required for the execution of the Sub Contract works.	Rs 2,000.00
3	Failure to wear personal protective equipment (P.P.E.) eg. Safety helmets, safety boots, goggles etc. respirator, ear plugs, safety belts which shall include failure to anchor belt to a secure structure. Where any site operation requires the use of PPE then all workmen must use the required PPE eg. grinding, welding, burning, unloading hazardous materials etc.	1) Rs.500.00 per worker when lack of enforcement of the usage of P.P.E. by the Sub Contractor is observed by the employer. OR 2) Rs.1, 000.00 where issuance of the required P.P.E. by the Sub Contractor equipment is not carried out.
4	Failure to attend general safety induction course conducted by the employer / the employer.	1) Rs.300.00 per worker for not attending the course; and 2) Workers to attend course within 2 working days or be dismissed.
5.	Failure to attend a notified site safety meeting.	Rs 2,000.00
6.	Failure to submit, within the specified time to the employer, safety supervisor reports which shall include other relevant statutory reports made under the F&IUO Cap. 59, tool box briefing records, weekly labour return, issuance of personal protective equipment records, safety data sheets of toxic and harmful materials and others related certificates.	Rs 2,000.00
7.	Failure to submit a written report for an accident and/or other dangerous occurrence, to the employer within 24 hours of its occurrence.	Rs 2,000.00
8.	Failure to carry out within the specified time the necessary improvement action against any notified safety violation.	Rs 5,000.00
9.	Damaged to or misuse of the employer's property.	1) Rs.2,000.00; and 2) In addition the Sub Contractor to pay for the cost of items damaged.
10.	Failure to maintain work area, facility storage and preparation yard, office premises and workers changing and	1) Rs 2,000.00; and

	rest area in a clean and orderly state and free from health and fire hazards.	2) In addition the Sub Contractor shall clean up the disorderly and untidy areas within 3 days.
11.	Obstruction of passageways, entrances, doorways, stairs, access to firefighting equipment etc. and/or the erection unsafe accesses and crossings.	Rs 2,000.00
12.	Use of equipment that has not been examined by an approved person as required under the Factories and Industries Undertaking Ordinance and its related regulations.	Rs 5,000.00
13.	Using defective or uncertified slings for lifting operations.	Rs 5,000.00
14.	Executing unsafe hoisting of materials and include unsafe use of lifting appliances.	Rs 5,000.00
15.	Erecting and/or using unsafe or unstable, scaffolds, working platforms and temporary structures.	Rs 5,000.00
16.	Failure to provide and use proper working platforms and safe means of access to the work place, where work is required to be carried beyond a person's normal reach.	1) 5,000.00; and 2) The worker to be immediately banned from the site if an individual receives 2 violations.
17.	Allowing workers to occupy or work on unguarded elevated platforms, floor edges and without adopting adequate safety measures against the risk of a person falling from height.	Rs 7,000.00
18.	Not providing safety barricades/barriers to hazardous floor edge openings, gaps and shafts.	Rs 7,000.00
19.	Rendering scaffold or working platform unsafe by tampering/alternation.	1) Rs 5,000.00 and 2) All rectification costs to be borne by the subcontractor.
20.	Placing of heavy items unsafely on scaffold or working platform.	Rs 2,000.00
21.	Throwing or allowing objects to drop from heights.	1) Rs 7,000.00 and 2) Workers to be immediately banned from the site.
22.	Stacking or leaving materials including work in progress articles and tools in unstable condition and or along floor edges such they are likely to endanger workers.	Rs 7,000.00
23.	Failure to effectively cordon off guard and warn other workers from entering into the danger areas when they are likely to be affected by falling materials from the Sub Contract Work.	Rs 7,000.00
24.	Violating the permit to work system for example, but not limited to:- a) Look out procedures b) Confined space entry permits c) Other hazardous work Eg. hot work (in the event of failure to comply)	1) Rs 5,000.00; and 2) Workers to be immediately banned from the site.

25.	Dismantling and rendering any safety guards or protective features of any part of a machine or any part of a building structure to the extent that such guards and protective features are not operational or are incapable of providing the necessary protection for its design and purpose.	Rs 7,000.00
26.	Adopting unsafe tapping, connections, and termination of electrical lines and including the use of defective electrical fittings, power cables and electrical tools. Allowing cables / equipment to be submerged in water.	Rs 2,000.00
27.	Using any defective or unsafe equipment.	Rs 2,000.00
28.	Unauthorized use of fire equipment provided for emergency purposes.	Rs 2,000.00
29.	Failure to comply with an order issued by the employer's Construction manager, site agent, site safety supervisor safety officer and environmental officers in regard to safety/environmental matters.	Upto Rs 7,000.00
30.	Threatening safety/personnel misbehavior, fighting or intentional causing hurt to others.	1) Rs 7,000.00 & to be blacklisted and 2) Worker to be banned from the site and a report will be made to the police.
31.	Other safety/environmental violations which are in breach of safety or environmental ordinances and regulations.	Rs 1,000.00
32.	Failure to wear safety harness and anchor to a secure structure, whilst working on a gondola.	Rs 7,000.00 and worker to be banned from site immediately.
33.	Tampering with the safety features of a gondola / winches ./ scaffolding	Rs 7,000.00 and worker to be banned from site immediately.
34.	Access and egress to gondola other than at a designated landing.	Rs7,000.00 and worker to be banned from site immediately.
35.	Failure to provide valid certificates for lifting appliances and accessories including any lifting appliances / accessories on vehicles delivering goods to the site, or for plant being used on site.	Rs5,000.00
36.	Failure to provide voltage-reducing devices on welding machines.	Rs 2,000.00
37.	Obstruction of passageways, entrances, doorways, stairs, access to firefighting equipment etc. and/or the erection unsafe accesses and crossings.	Rs 2,000.00
38.	Failure to attend the Morning Site Assembly Meetings.	Rs 500.00
39.	Failure to tidy up the Sub contract works after each day work	Rs1,200.00
40.	Failure to attend the Site Safety Walk	Rs 1,000.00
41.	Failure to provide Site Safety Supervisor/Representative as per the requirement of the latest ordinance.	Rs 2,000.00
42.	Failure to use stairs in lieu of lifts	Rs 500 per event

T FOR STATUTORY COMPLIANCE REGISTERS
TO BE MAINTAINED BY A CONTRACTOR

Note:

1. ***The following are the mandatory list of registers that need to be maintained as per the statutory requirements (the items may vary as per Rules of the State in which project is being executed)***
2. ***The registers need to be maintained only for the employees deployed at the site***
3. ***One set of photocopies of all the documents to be submitted at the time of audit***
4. ***The following documents to be maintained for the period commencement of contract to completion of contract***

1. Contract License
2. Form VI-A commencement of Contract intimation to the department
3. Muster Roll
4. Wage Register
5. Register of overtime
6. Register of Advance
7. Register of Fine
8. Register of Workman Employed by the Contractor
9. Employment Card
10. In & out timings register
11. PF registration copy
12. PF Challans
13. PF contribution details for the employees deployed at the site (from PF web portal)
14. ESIC Challans
15. ESIC Temp card copy of for the employees deployed at the site
16. ESIC Contribution details for the employees deployed at the site (from ESIC web portal)
17. Vendors monthly compliance undertaking (as per the excel sheet attached)

VENDOR MONTHLY COMPLAINCES FORMAT

Name & Address of the Vendor	Name of the Principal Employer & Address	Month	
		ESIC.Reg. No.	
		EPF . Reg No.	
:		Professional Tax. Re	

SL No.	Name of the employee	Father's Name	Employee's PF Number	Employee's ESI Number	No.of days worked during the month	Basic wage	DA	HRA &Other Allowance	Over time Hour	Overtime wages	Gross	DeductionOf EPF	Deduction ESIC	Deduction PT	Deduction LWF	NET Amount

We hereby Certify that the above mentioned employees PF/ESIC has been remitted with the respective department against challan attached for the Month of..... .. We hereby affirm that we will be liable to pay any financial liabilities in case of such liabilities arise in future in regard to the documents submitted by us.

18. Labor Welfare Fund contribution at Haryana
19. Cess payment Receipt (As per the BOCW registration obtained by Client)

Any other document/Register as may be required per the prevailing Statutory/Legal requirements of the State

5. ***The attached documents to be maintained monthly labour compliance audit will be conducted. Final bill payment will be released based on labour compliance audit clearance.***

AUDIT CHECKLIST						
SITE NAME :		DATE OF AUDIT				
LOCATION:		PLACE OF AUDIT				
NAME OF CONTRACTOR		NAME OF AUDITOR				
ADDRESS:		No of Employees				
		Nature OF Work				
NAME OF REPRESENTATIVE						
CONTACT NUMBER/E-Mail		Contract Period				

Sl No	Act	Compliance check	Responsible	Compliant/Non Compliant	April	
1	BOCW Act	Whether Registration under BOCW Act has been obtained				
2	BOCW Act	Whether Non-Compliantttice of Commencement/completion has been filed				
3	BOCW Act	Whether Cess has been paid				
4	BOCW Act	Whether Annual Return has been filed				
5	BOCW Act	Whether Certificate of Registration has been displayed at site				
6	BOCW Act	Whether Abstract has been displayed at site				
7	BOCW Act	Whether information related to Working Hours, Rate of Wages, Rest Day, Payment date has been displayed at site				
8	BOCW Act	Whether Name of Inspector with Address has been displayed.				
9	BOCW Act	Whether place and time of disbursement of wages has been displayed at site				
10	BOCW Act	Whether overtime has been paid to employees @ double rate of wages				
11	BOCW Act	Whether any employee worked more than 9 hours on any day or more than 48 hours in a week				
12	BOCW Act	Whether interval for rest has been provided to employees on every five hours of working.				
13	BOCW Act	Whether any employee's spread hours are more than 12 hours in a day (including interval for rest)				
14	BOCW / CLRA / MW / PW Act	Whether "Register of Building workers employed by the Employer" has been maintained.				
15	BOCW / CLRA / MW / PW Act	Whether Muster Roll has been maintained				
16	BOCW / CLRA / MW / PW Act	Whether Rigister Of Wages has been maintained				
17	BOCW / CLRA / MW / PW Act	Whether Form Of Register Of Wages-Cum-Muster-Roll has been maintained				
18	BOCW / CLRA / MW / PW Act	Whether Register Of Deductions For Damages Or Loss has been maintained				
19	BOCW / CLRA / MW / PW Act	Whether Register Of Fines has been maintained				
20	BOCW / CLRA / MW / PW Act	Whether Register Of Advances has been maintained				
21	BOCW / CLRA / MW / PW Act	Whether Register Of Overtime has been maintained				
22	BOCW / CLRA / MW / PW Act	Whether Wage Book has been maintained				
23	BOCW / CLRA / MW / PW Act	Whether Service Certificate has been issued to workers				
24	BOCW / CLRA / MW / PW Act	Whether Identity Card Issued To Building Workers				
25	BOCW Act	Whether any accident or dangerous occurred at site				
26	BOCW Act	Whether Register of accident and dangerous occurrence has been maintained				
27	BOCW Act	Whether rest day provided on every six days of working				
28	BOCW Act	In case employee worked on rest day, compensatory off has been provided within 5 days and given overtime for working on rest day.				
29	BOCW Act	In case any worker completed more than 90 days, whether same worker has been registered as beneficiary				
30	BOCW Act	Whether employee and employer contribution has been paid to department on monthly basis				
31	BOCW Act	whether monthly return related to contribution has been filed with department				
32	BOCW Act	Whether Register related to Beneficiary workers has been maintained				
33	CLRA Act	Whether CLRA Applicable (if project period is more than 90 days)				
34	CLRA Act	Labour License and Validity				
35	CLRA Act	Form-VI A for Commincement and Complitions				
36	Minimum Wages Act	Whether applicable minimum wages has been paid to workers				
37	ESIC Act	Whether ESIC is applicable on Establishment				
38	ESIC Act	If applicable then whether ESIC has been deducted from employees salary as per provisions				
39	ESIC Act	Whether ESIC challan has been generated and ECR is matching with deduction as per Wage Register				
40	ESIC Act	Whether ESIC Contribution has been paid on time.				
41	ESIC Act	Whether ESIC contribution has been paid in the local ESIC sub Code or mapped to another location.				
42	PF ACT	Whether PF is applicable on Establishment				
43	PF ACT	Whether company deposit PF with RPFC or have own PF Trust.				
44	PF ACT	If applicable then whether PF has been deducted from employees salary as per provisions				
45	PF ACT	Whether PF challan has been generated and ECR is matching with deduction as per Wage Register				
46	PF ACT	Whether PF Contribution has been paid on time.				
47	Labour Welfare Fund Act	If applicable, whether LWF has been deducted as per applicable slabs				
48	Labour Welfare Fund Act	Whether LWF has been paid on time.				
49	Professional Tax Act	If completed then whether PT has been deducted and paid				
50	Professional Tax Act	If applicable Monthly / Yearly Return				
51	Equal Remuneration Act	Whether wages has been paid to male and female employees at the same rate and there is no difference based on the gender.				
52	Child Labour Act	Whether employed any child labour who is below 15 years of age.				
53	Child Labour Act	Whether employed any young person who has completed 15 years of age but below 18 years of age				
54	Inter State Migrant Labour Act	Whether five or more inter state migrant workers employed				
55	Inter State Migrant Labour Act	Whether Contractor has applied for Licence is this Act is applicable				
56	Inter State Migrant Labour Act	Whether Pass book has been issued to workers as per provision of this Act (If applicable)				
57	Inter State Migrant Labour Act	Whether periodic return has been filed under this Act				
58	Employee Compensation Act	Whether employees who are not covered under ESIC Act, are covered under employee compensation policy.				

Occupational Health, Safety and Environment (OHSE) Requirements

29) INTRODUCTION

This document illustrates the principle requirement of CBRE South Asia Pvt Ltd, on Occupational Health, Safety and environment associated with the Contractor. The CONTRACTOR, his Subs and DC's, shall comply with the safety precautions, protective measures, housekeeping requirements, etc. The Project Managers / Architects with due intimation shall have the right to stop the work at site, if in his opinion proceeding with the work will lead to an unsafe and dangerous condition. The CONTRACTOR shall get the unsafe condition removed and also provide protective equipment. The CONTRACTOR shall ensure that all workmen are aware about the nature of risk involved in their work and have adequate knowledge for carrying out their work safely.

The instructions issued herein are indicative and not exhaustive. Therefore the CONTRACTOR shall be responsible to ensure that adequate safety measures have been adopted in the course of execution of the contract in accordance with safety standards / statutory regulations, as applicable.

The CONTRACTOR shall be held responsible for non-compliance if any of the safety measures and delays, implications, injuries, fatalities and compensation arising out of such situations or incidents.

30) OHSE TARGET AND GOALS

- Zero recordable injuries
- Zero reportable environmental incidents
- 100% adherence to usage of proper PPE at work.
- 100% incident recording and reporting

31) DUTIES/RESPONSIBILITIES OF OHSE COORDINATOR/CONTRACTOR

1) Safety person shall wear green colored helmet while at work.
2) The project Manager shall be responsible for implementing and maintaining safety at work site and the EHS representatives shall advise , assist & provide training and create awareness to all about safe practices at work place Responsibility lies on all team members of the project to achieve the following goals
    a) 100% implementation of safety rules and regulations.
    b) Checking work areas , making housekeeping inspections
    c) Keeping records of all unsafe conditions found and corrective action taken.
    d) Enabling members to use PPE
    e) Conducting safety committee meeting on a periodic basis to discuss on the safety non compliances.

32) SAFETY ORGANISATION/SAFETY COMMITTEE

- A safety committee should be formed at site. The members should be selected from Client, PMC and the Contractor. The Project Manager will be the Chairperson of the committee.
- Meeting should be conducted on a periodic basis with all the safety committee members. MOM should be circulated among the members. Project managers shall ensure the compliance of the issues raised in the meetings as per the target dates mentioned in MOM

33) TRAFFIC

vi. The CONTRACTOR shall organize operations by taking approval to use the existing roads, routes / access, if any, from Client.

vii. The CONTRACTOR shall exercise full care to ensure that no damage is caused by him or workmen, during the operation, to the existing water supply, sewerage, power or telecommunication lines or any other services or works. The CONTRACTOR shall provide and erect before construction, substantial barricades, guardrails, and warning signs. He shall furnish, place and maintain adequate warning lights, signals, etc., as required by Project Managers / Architects.

viii. Pedestrian walkways should be segregated from the path of vehicle movement.

ix. Speed limit sign boards should be displayed at site.

x. Speed breakers must be provided so as to reduce the over speed of vehicles.

34) SAFE MEANS OF ACCESS/ EGRESS

i. Adequate and safe means of access and exit shall be provided for all work places, at all elevations.

ii. Suitable scaffolds shall be provided for workmen for all works that cannot safely be done from the ground, or from solid construction except such short duration work as can be done safely from ladders. Ladder shall be of rigid construction having sufficient strength for the intended loads and made of metal and all ladders shall be maintained well for safe working condition. Suitable footholds and handholds shall be provided on the ladder. The ladder shall be given an inclination not steeper than 1 in 4 (1 horizontal and 4 vertical / at an angle of 75 degree to the wall)

iii. Access/Egress should be free of obstructions and from any slippery substances like oil, water or grease.

iv. The landing space for the access/egress should be a leveled surface.

35) SCAFFOLDS

x. Scaffolding or staging more than 3.5m above the ground or floor, swung or suspended from an overhead support or erected with stationary support shall have a standard guard-rail properly attached, bolted, braced or otherwise secured at least 1m high above the floor or platform of such scaffolding or staging. The guardrail shall extend along the entire exposed length of the scaffolding with only such opening as may be necessary for the delivery of materials. Work platforms must have hand rails at 1m height and Mid Rails at 0.5m level. Such scaffolding or staging shall be so fastened as to prevent it from swaying from the building or structure. Scaffolding and ladder shall conform to relevant IS specification (IS 3696-1966). TIMBER/BAMBOO SCAFFOLDING SHALL NOT BE USED.

xi. Working platforms of scaffolds shall have toe boards at least 15cm in ht. to prevent materials from falling down.

xii. A sketch of the scaffolding proposed to be used shall be prepared prior to start of erection of scaffolding. Authorized scaffolder shall examine all scaffolds before using.

xiii. Working platform, gangways and stairways shall be so constructed that they shall not sag unduly or unequally and if the height of the platform or gangway or stairway is more than 3.5m above ground level or floor level, they shall be closely boarded, shall have adequate width for easy movement of persons and materials and shall be suitably guarded.

xiv. The planks/jalley used for working platform shall not project beyond the end supports to a distance exceeding four times the thickness of the planks used. The planks/jalleys shall be rigidly

tied at both ends to prevent sliding and slippage. The thickness of the planks/jalleys shall be adequate to take load of men and materials and shall not collapse.

xv. Every opening in the floor of a building or in a working platform shall be provided with suitable means to prevent fall of persons or materials by providing suitable barricades, fencing or railing, the minimum height of which shall be 1m, along with 15cm high sheet obstruction at floor level along the railing.

xvi. Safe means of access shall be provided to all working platforms and other elevated working places. Every ladder shall be securely fixed. No single portable ladder shall be over 9m in length. For ladders, up to 3m in length the width between side rails in the ladders shall in no case be less than 300mm. For longer ladders, this width shall be increased by at least 20mm for each additional meter of length. Step spacing shall be uniform and shall not exceed 300mm.

xvii. Adequate precautions shall be taken to prevent danger from electrical lines and equipment. No scaffolding, ladder, working platform, gangway runs, etc. shall exist within 3 meters of any un-insulated electric wire. Whenever electric power and lighting cables are required to run through (pass on) the scaffolding or electrical equipment are used, such scaffolding structures shall have minimum two earth connections with earth continuity conforming to IS code of practice.

xviii. Bottom of the scaffold should be provided with base plates. Proper cross bracings should be given for supporting the scaffold.

36) DEMOLITION

vii. Before any demolition work is commenced and also during the progress of the work:

viii. All roads and open area adjacent to the work site shall either be closed or suitably protected. Appropriate warning signs shall be displayed for cautioning approaching persons.

ix. Before demolition operations begin, the CONTRACTOR shall ensure that job safety analysis is prepared and take approval from CBRE/Client & the power on all electric service lines is shut off and the lines cut or disconnected at or outside the demolition site. If it is necessary to maintain electric power during demolition operation, the required service lines shall be adequately protected against damage with a licensed electrician only All Persons including those handling heavy materials/equipment shall also wear all basic PPEs at the time of entire operations

x. Contractor shall submit waste management plan and only after approval by client the demolishing activity will be carried out. No floor, roof or other part of the building shall be overloaded with debris or materials as to render it unsafe. In case of any noncompliance of the waste management plan contractor will be liable for disciplinary procedures

xi. Entries to the demolition area shall be restricted to authorized persons only.

xii. Adequate caution signage's, effective gate management shall be ensure by the contractor while carrying out the job.

37) PERSONAL PROTECTIVE EQUIPMENT

xi. All necessary personal protective equipment shall be kept available for the use of the persons employed on the site and maintained in a condition suitable for immediate use. Also the CONTRACTOR shall take adequate steps to ensure proper use of equipment by those concerned. The personal protective equipment to be provided by the CONTRACTOR is detailed below.

xii. All persons employed at the construction site shall use safety helmets. No worker will be allowed entry to the site without safety helmets and ID cards.

xiii. Workers employed on mixing asphaltic materials, cements and lime mortars shall use protective goggles, protective footwear and hand gloves.

xiv. Persons engaged in welding and gas-cutting works shall use suitable aprons, leather gloves and welding face shields. The persons who assist the welders shall use suitable goggles. Protective goggles shall be worn while chipping and grinding.

xv. Stonebreakers shall use protective Goggle, Face shield & Nose Mask. They shall be seated at sufficiently safe intervals of distance.

xvi. Persons engaged in or assisting in shot blasting operations and cleaning the blasting chamber shall use suitable gauntlets, overalls, dust-proof goggles, boots and protective hood supplied with fresh air at the minimum rate of 9m/hr.

xvii. All persons working at height more than 2.5m above ground or floor and exposed to risk of falling down shall use full body safety harness, unless otherwise protected by cages, guard railings, etc. In places where the use of safety belts is impractical, suitable net of adequate strength fastened to substantial supports shall be employed.

xviii. Protective nets shall be provided at places where men are working at heights to safe guard against fall of men or material including debris. Contractor shall provide head protection where height works are carried out where workers movement cannot be restricted or re directed.

xix. When workers are employed in sewers and inside manholes that are in use, the CONTRACTOR shall ensure that the manholes are opened and are adequately ventilated at least for an hour. After it has been well ventilated, the atmosphere inside the space shall be checked for the presence of any toxic gas or oxygen deficiency and recorded in the register before the workers are allowed to get into the manholes. The manholes opened shall be cordoned off with suitable railing and provided with warning signals or caution boards to prevent accidents. Contractor shall submit the method statement and job safety analysis and the work can be allowed to carry out only after approval of CBRE/Client. Confined space activities shall be carried out by the contractor after obtaining work permit from the client. Night work in such areas will be allowed only if all the contents of the work permit are compiled and close monitoring is ensured.

xx. In case the CONTRACTOR fails to supply the safety equipments, the owner may purchase the same and debit the Price of the equipments and incidentals, to the CONTRACTOR.

S #	Name of Safety Equipment/ Accessories
1	Safety Helmet (Design for carrying load)
2	Regular Safety Helmet / Hard Hat
3	Industrial Safety Shoes
4	Safety Gum Boot
5	Safety Goggles
6	Safety Gloves (Cotton)
7	Safety Gloves (Electrical)
8	Safety Gloves (Leather)

9	Safety Gloves (Rubber)
10	Particulate Respirator
11	Ear plug
12	Full Body Harness
13	Safety Nets -12mm; 10m x 3m
14	Safety Rope
15	Single Sided Barricading Tape Red & White / Yellow & Black
16	Laminated Safety Poster 18" x 12" [1 For 100 Sq. M]
17	Fire Extingisher-CO2-4.5 Kg
18	Fire Extingisher-ABC-5Kg
19	Sand Bucket
20	Fire Blanket (1mtrX1mtr)
21	Aluminum Ladder

38) PAINTING

xii. The CONTRACTOR shall not employ women on the work of painting with products containing lead in any form. Only men above the age of 18 years shall be employed on the work with lead paint. The following precautions shall be taken during the work:

xiii. Supply air respirators for use by the workers when paint is applied in the form of spray, or a surface having lead paint is scraped.

xiv. The workmen shall use overalls and adequate facilities shall be provided to enable the painters to wash at the cessation of work.

xv. All painting jobs, especially those in which lead paints are used shall be kept under industrial hygiene surveillance.

xvi. Smoking, open flames or sources of ignition shall not be allowed in places where paints and other flammable substances are stored, mixed or used. A caution board, with the instructions written in national/regional language, "SMOKING - STRICTLY PROHIBITED" shall be displayed in the vicinity where painting is in progress or where paints are stored. Symbols shall also be used for caution boards.

xvii. Suitable fire extinguishers/sand buckets shall be kept available at places where flammable paints are stored, handled or used.

xviii. When painting work is done in a closed room or in a confined space, adequate ventilation shall be provided. If adequate ventilation cannot be provided, workers shall wear suitable respirators.

xix. Epoxy resins and their formulations used for painting shall not be allowed to come in contact with the skin. The workers shall use protective gloves or suitable barrier creams.

xx. Adequate ventilation shall be provided especially when working with hot resin mixes.

xxi. Increased personal hygiene shall be practiced to control inadvertent contact with the resin and eliminate its effects.

xxii. Workers shall thoroughly wash hands and feet before leaving the work. Work clothes shall be changed and laundered frequently.

39) LIFTING MACHINES AND TACKLES

xii. Use of lifting machines and tackles including their attachments, anchorage and supports shall confirm to the following standards or conditions:

xiii. Lifting machines and tackles shall be of good mechanical construction, sound material and adequate strength and free from any defects and shall be kept in good repair and in good working order.

xiv. Every rope used in hoisting or lowering materials or as a means of suspension shall be of good quality and adequate strength and free from any defect.

xv. Every crane operator or lifting appliance operator shall be properly qualified. No person under the age of 21 years shall be in charge of any hoisting machine or give signal to operator of such machine.

xvi. In case of every lifting machine (and of every chain, ring, hook, shackle, swivel and pulley block used in hoisting or as means of suspension) the safe working load shall be ascertained and clearly marked. In case of a lifting machine having a variable safe working load, each safe working load and the conditions under which it is applicable shall be clearly indicated. No part of any machine or any gear referred to above in this paragraph shall be loaded beyond the safe working load except for the purpose of testing. The safety engineer shall approve this.

xvii. The safety engineer shall note the safe working load. Regarding other machines, the CONTRACTOR shall notify the safe working load of the machine to the safety engineer, whenever he brings any machinery to site of work and gets it verified by the safety engineer.

xviii. Thorough inspection and load testing of lifting machines and tackles shall be done by a competent person at least once every 6 months and records of such inspection and testing shall be maintained.

xix. Motors, gearing transmission, couplings, belts, and chain drives and other moving parts of hoisting appliances shall be provided with adequate safeguards. Hoisting appliances shall be provided with such means as will reduce to the minimum the risk of any part of a suspended load becoming accidentally displaced or lowered. Barricades shall be erected around the place of hoisting the equipment(s).

xx. Load shall be never allowed to swing during lifting.

xxi. Tires should be inspected for cuts, tears, breaks and proper inflation.

xxii. Copy of third party test certificate for slings should be submitted to the safety dept.

40) WELDING AND GAS CUTTING

xv. Welding and gas cutting operations shall be done by qualified and authorized persons and as per IS specifications and code of practice.

xvi. Welding and gas cutting shall not be carried out in places where flammable or combustible materials are kept and where there is danger of explosion due to presence of gaseous mixtures.

xvii. Welding and gas cutting equipment including hoses and cables shall be maintained in good condition.

xviii. Barriers shall be erected to protect other persons from harmful rays from the work. When welding or gas cutting is done in elevated positions, precautions shall be taken to prevent sparks or hot metal falling on persons or flammable materials.

xix. Suitable type of protective clothing consisting of fire resistant gauntlet gloves, leggings, boots and aprons shall be provided to workers as protection from heat and hot metal splashes. Welding shields with filter glasses of appropriate shade shall be worn as face protection.

xx. Adequate ventilation shall be provided while welding in confined space or while brazing, cutting or welding zinc, brass, and bronze, galvanized or lead coated materials.

xxi. Welding and gas cutting shall not be done on drums, barrels, tanks or other containers unless they have been emptied, cleaned thoroughly and it is made certain that no flammable material is present.

xxii. Fire extinguisher shall be available near the location of welding operations. Fire safety permit shall be obtained for working at vulnerable areas and operating areas before flame cutting/welding is taken up.

xxiii. For electric (Arc) welding the following additional safety precautions shall be taken:

xxiv. When electrical welding is undertaken near pipelines carrying flammables, such pipelines shall not be used as part of earth conductor, but a separate earth conductor shall be connected to the machine directly from the job.

xxv. Personnel contact with the electrode or other live parts of electric welding equipment shall be avoided.

xxvi. Extreme caution shall be exercised to prevent accidental contact of electrodes with ground.

xxvii. The welding cable shall not be allowed to get entangled with power cables. It shall be ensured that movement of materials does not damage the cables.

xxviii. For gas cutting the following safety precaution should be taken:

a) Four numbers of Flash back arrestors should be present as two for cylinders and two for torch

b) The sign of flashback will be a loud crack, followed by a puff of carbon from the welding tip when relighting.

c) The consequence of not arresting a flashback can range from bursting of hose to a bombing explosion of cylinder.

d) The least damage a flashback can do is to deposit carbon in torch valves which can affect their operation and can lead to more serious flashbacks.

e) Never join acetylene hoses with copper tube or copper wire since they may produce an explosive mixture.

f) Never set the acetylene cylinder outlet gauge pressure more than 100kPa

g) Acetylene smells like onion or garlic, which helps in detecting leak.

41) GRINDING

vi. All portable grinders shall be used only with their wheel guards in position to reduce the danger from flying fragments should the wheel break during the use.

vii. Grinding wheels of specified diameter only shall be used on a grinder – portable or pedestal - in order not to exceed the prescribed peripheral speed.

viii. Goggles shall be used during grinding operation.

ix. Guard in the wheel shall never be removed during work.

x. Grinding the material on the sides of the wheel can damage the wheel.

42) DRILLS:

a. Care should be taken not to drill too large hole at a time.

b. Drilling of large holes should be done in stages by using small bore drills and gradually increasing the size as necessary.

c. Bit should be properly tightened to the chuck.

d. Auxiliary handle should be used for continuous work.

e. Do not drill with one hand while holding the material with other.

f. Do not raise or lower the drill by its power cord.

43) HOUSE KEEPING

vi. The CONTRACTOR shall at all-time keep his work site, site office and surroundings clean and tidy from rubbish, scrap, surplus materials and unwanted tools and equipment.

vii. Welding and other electrical cables shall be so routed as to allow safe traffic by all concerned.

viii. No materials on any of the sites of work shall be so stacked or placed as to cause danger or inconvenience to any person or the public.

ix. At the completion of the work, the CONTRACTOR shall ensure removal from the work premises all scaffoldings, surplus materials, rubbish and all huts and sanitary arrangements used/installed for workmen on the site.

x. Any rubbish materials should not be burnt at site.

44) ALCOHOL and DRUGS

1) The contractor shall ensure at all times that no Labour is working under the influence of alcohol/drugs. If found they should be suspended from the site immediately.

2) Smoking at worksite by any member is also prohibited.

45) FIRE SAFETY

vi. All necessary precautions shall be taken to prevent outbreak of fires at the construction site. Adequate provisions shall be made to extinguish fires, should they still break out.

vii. Quantities of combustible materials like timber, bamboo, coal, paints, etc. shall be the minimum required in order to avoid unnecessary accumulation of combustibles at site.

viii. Containers of paints, thinners and allied materials shall be stored in a separate room, which shall be well ventilated, and free from excessive heat, sparks, flame or direct rays of the sun. The containers of paint shall be kept covered and properly fitted with lid and shall not be kept open except while using.

ix. Fire extinguishers shall be located at the construction site at appropriate places.

x. Adequate number of workmen shall be given education and training in firefighting and extinguishing methods.

46) WORK IN RADIATION AREA

The CONTRACTOR shall follow the stipulated procedure regarding work in the radiation are and other works related with radiography.

47) WORK IN AND AROUND WATER BODIES

When the work is done near any places where there is risk of drowning, all necessary rescue equipment such a life buoys and life jackets shall be provided and kept ready for use and all necessary steps taken for prompt rescue of any person in danger and adequate provision shall be made for prompt first-aid treatment of all injuries likely to be sustained during the course of the work. Persons who do not know swimming shall not be engaged alone for any work where risk of drowning exists. Sufficient number of life buoys or life jackets shall be provided.

48) MEDICAL FACILITIES

v. The CONTRACTOR shall arrange for medical aid and treatment for his staff and workers engaged on the work site including the first-aid facilities if they are not available at the project site.

vi. First-aid appliance including sterilized dressing, cotton wool and antiseptic cream shall be made available at readily accessible places at every work site. These shall be maintained in good order under the charge of a responsible person.

vii. Each first aid box shall contain the following equipment and adequate to the number of workers

(As per The BOCW Act 1996)

1) A sufficient number of eye wash bottles filled with distilled water or suitable liquid clearly Indicated by a distinctive sign which shall be visible at all times.
2). 4% xylocaine eye drops, and boric acid eye drops and soda bicarbonate eye drops.
3). 24 – Small sterilized dressings.
4). 12 – Medium size sterilized dressings.
5). 12 – Large size sterilized dressings.
6). 12 – Large size sterilized burn dressings.
7). 12 – (Fifteen cms) packets of sterilized cotton wool.
8). (200 ml) bottle of centrimide solution (1%) or suitable antiseptic solution.
9). One (200 ml) bottle of Mercurochrome solution (2%) solution in water.
10). One (120 ml) bottle of salvolatile having the doses and mode of administration indicated on the label.
11). One pair of Scissors.
12). One roll of adhesive plaster (6 cm x 1metre).
13). Two rolls of adhesive plaster (2 cm x1 meter).
14). 12 pieces of sterilized eye pads in separate sealed packets.
15). A bottle containing 100 tablets (each of 325 mg) of aspirin or any other analgesic.
16). 12 – roller bandages 10 cms wide.
17). 12 – Roller bandages of 5 cms wide.
18). One tourniquet.
19). A supply of suitable splints.
20). 3 packets of safety pins.

21). Kidney tray.
22). A snake bite lancet.
23). One (30 ml) bottle containing potassium permanganate crystals.
24). Copy of First-aid leaflet issued by Director General.
25). 6 – Triangular bandages.
26). 2 pairs of suitable, sterilized, latex hand gloves.

viii. At large work places where hospital facilities are not available within easy reach of the work, first-aid posts shall be established. Ambulance availability shall be identified during the entire period of work for attending to injury cases.

49) SAFETY COORDINATOR

The CONTRACTOR shall have a Safety Officer or a Supervisor to be designated as a Safety Coordinator in order to specifically look into the implementation of various safety requirements of the site work. The person thus designated will in general co-ordinate on matters of safety and in particular ensure that the Safety Manual is complied with. His name shall be displayed on the Notice Board at a prominent place at the work site.

50) LABOUR WELFARE

5) The contractor shall provide separate urinals/toilets for male and female workers as per the guidelines given in BOCW Act-1996
6) Drinking water should be made available for the workers at site The drinking water should be tested for every six months and the report should be submitted by the contractor to CBRE
7) In every workplace wherein not less than 100 workers are ordinarily employed, the contractor shall provide an adequate canteen for the workers.
8) The contractor shall provide temporary accommodation to all workers. These accommodations shall have bathing and washing & good sanitation facilities including rest rooms.

51) REPORTING OF ACCIDENT

iv. All accident leading to property damage and/or personnel injuries shall be reported to the concerned authorities' viz. Insurance Co. Police, Head Office, Regional Office, etc.

v. The CONTRACTOR shall also submit a monthly statement of accidents to the Project Managers / Architects by every month showing details of accident, nature of injury including disability, days lost, treatment provided, etc., and the extent of property damage.

vi. The statutory safety records required to be maintained by the CONTRACTOR at the site are

a. Accident Register
b. Dangerous Hazards Register
c. First Aid Register
d. Register for the inspection notes by government officers.
e. Copies of all accident reports and hazard reports filed to the prescribed authorities.
f. Incident/Near-miss registers

52) PUBLIC PROTECTION

The CONTRACTOR shall make all necessary provisions to protect the public. He shall be held responsible for defense of every action of other proceedings at law that may be brought by any person for injury sustained owing to neglect of any precaution required to taken to protect the public.

53) OTHER STATUTORY PROVISIONS

All operations involving the transport, handling, storage and use of explosive shall be as per the standing instructions and conform with the latest Indian Explosives Act and the explosives Rules. Handling, transport, storage and use of compressed gas cylinders and pressure vessels shall conform to the latest Gas Cylinder Rules and Static and Mobile Pressure Vessels (Unfired) Rules. In addition, The Indian Electricity Act and Indian Electricity Rules - latest, the Atomic Energy Act, the Radiation Protection Rules - latest, Radiation Protection Manual of Nuclear Facilities and the Atomic Energy (Factories) Rules – latest, and various latest rules and Act related to mining shall also be strictly complied with.

54) GUIDELINES AND GENERAL PROCEDURES FOR SUPPLY AND USE OF ELECTRICITY AT SITE

1. **GENERAL**

Following safety requirements shall be complied with before the CONTRACTOR uses the power supply.

11. The CONTRACTOR shall submit a list of licensed electrical staff to be posted at Site.
12. It shall be the responsibility of the CONTRACTOR to provide and maintain complete installation on the load side of the supply point with regard to the safety requirements at Site. All cabling and installation shall comply with the appropriate latest statutory requirements given below and shall be subject to approval of the Engineer-in-Charge:
    f. Indian Electricity Act.
    g. Electricity (Supply) Act.
    h. Indian Electricity Rules.
    i. National Electricity Code.
    j. Other relevant rules of Local Bodies and Electricity Boards.
13. The power supply shall be regulated as per the terms and conditions of the supply of the respective electricity boards.
14. Where distribution boards are located at different places the CONTRACTOR shall submit schematic drawing indicating all details like size of wires, Overhead and Cable feeders, earthing etc. The position and location of all equipment and switches shall be given.
15. The CONTRACTOR shall make his own arrangement for main earth electrode and tapings thereof. The existing earth points available at site can be used at the discretion of Client with prior permission. Method of earthing, installation and earth testing results shall conform to relevant I.S. Specifications (IS-3043).
16. All three phase equipments shall be provided with double earthing. All light fixtures and portable equipment shall be effectively earthed to main earthing.
17. All earth terminals shall be visible. No gas pipes and water pipes shall be used for earth connection. Neutral conductor shall not be treated as earth wire.
18. The CONTRACTOR shall not connect any additional load without prior permission of Client.

19. Joints in earthing conductors shall be avoided. Loop earthing of equipment shall not be allowed. However, tapings from an earth bus may be done.

20. The entire installation shall be subjected to the following tests before energizing of installation including portable equipment: -

    a. Insulation resistance test.

    b. Polarity test of switches.

    c. Earth continuity test.

    d. Earth electrode resistance.

    The test procedures and their results shall conform to relevant standards.

2. **INSTALLATION**

1. Only persons having valid wireman's license/competency certificate shall be employed for carrying out electrical work and repair of electrical equipment, installation and maintenance at site. A qualified licensed Supervisor shall supervise the job.

2. Electrical equipment and installations shall be installed and maintained as to prevent danger from contact with live conductors and to prevent fires originating from electrical causes like short circuits, overheating etc. Installation shall not cause any hindrance to movement of men and materials.

3. Materials for all electrical equipment shall be selected with regard to working voltage, load and working environment. Such equipment shall conform to the relevant standards.

4. The minimum clearance to be maintained for all overhead lines along roads and across roads shall be as per the statutory requirements.

5. Grounding conductor of wiring system shall be of copper or other corrosion-resistant material. An extra grounding connection shall be made in appliances/equipment where chances of electric shock are high.

6. Electric fuses and/or circuit breakers installed in equipment circuits for short circuit protection shall be of proper rating. It is also recommended that high rupturing capacity (HRC) fuses are used in all circuits. Earth Leakage Circuit Breaker (ELCB) /Residual Current Circuit Breaker (RCCB) of 30 mA sensitivity shall be provided in the circuits.

7. Wherever cables or wires are laid on poles, a guard wire of adequate size shall be run along the cables/wires and earthen effectively. Metallic poles as a general rule, shall be avoided and if used shall be earthen individually. Anti-climbing guards and danger notices shall be provided on poles. Each equipment shall be an individual isolating switch.

8. Wires and cables shall be properly supported and an approved method of fixing shall be adopted. Loose hanging of wires and cables shall be avoided. Lighting and power circuits shall be kept distinct and separate.

9. Reinforcement rods or any metallic part of structure shall not be used for supporting wires and cables, fixtures, equipment, earthing etc.

10. All cables and wires shall be adequately protected mechanically against damages. In case the cable is required to be laid underground, it shall be adequately protected by covering the same

with bricks, Plain Cement Concrete (PCC) tile or any other approved means and Cable marker shall be displayed as per local standards.

11. Using suitable cable glands shall properly terminate all armoured cables. Using cable lugs/sockets shall connect multi-stranded conductor cables. Cable lugs shall preferably be crimped. They shall be of proper size and shall correspond to the current rating and size of the cable. Twisted connections will not be allowed.

12. All cable glands, armoring and sheathing of electric cable, metal circuits and their fittings, metallic fittings and other non-current carrying parts of electrical equipment and apparatus shall be effectively grounded.

13. All the Distribution Boards, Switch Fuse Units, Bus bar chambers, ducts, cubicles etc. shall have MS enclosures and shall be dust, vermin and waterproof. The Distribution Boards, switches etc. shall be so fixed that they shall be easily accessible. Changes shall be done only after the approval of the Engineer-in-Charge.

14. The CONTRACTOR shall provide weather proof metal enclosures/covers for protection of the entire switchboard, equipment. Exposed live parts of all electrical circuits and equipment shall be enclosed permanently. Crane trolley wires and other conductor which cannot be completely insulated shall be placed such that they are inaccessible under normal working conditions.

15. Iron-clad industrial type plug & sockets with Cap are preferred for additional safety.

16. PDBs shall have emergency contact details, Signage, incomer details.

17. Isolating switches shall be provided close to equipment for easy disconnection of electrical equipment or conductors from the source of supply when repair or maintenance work has to be done on them.

18. In front of distribution boards a clear space of 90 cm shall be maintained in order to have easy access during an emergency.

19. Adequate working space shall be provided around electrical equipment, which requires adjustment or examination during operation.

20. As far as possible electrical switches shall be excluded from a place where there is danger of explosion. All electrical equipment such as motors, switches and lighting fittings installed in workroom where there is possibility of explosion hazard shall be explosion proof.

21. All connections to lighting fixtures, starters or other power supplies shall be provided with PVC insulate, PVC sheathed twin/three/four core wires to have better mechanical protection for preventing possible damage to equipment or injury to personnel. Taped joints shall not be allowed and the connections may be made in looping system. Electric starter of motors, Switches shall not be mounted on wooden boards. Only sheet steel mounting or iron framework shall be used.

22. All the lighting fixtures and lap holders shall be of good quality and in good condition. Badly repaired or broken holders, etc. shall not be used.

23. Only PVC insulated and PVC sheathed wires or armored PVC insulated and sheathed cables shall be used for external power supply connections of temporary nature. Weatherproof rubber wires shall not be used for any temporary power supply connections. Taped joints in the wires shall not be used.

24. The bulbs/lamps used for illumination and testing purpose shall have cover or guard to protect them from accidental breakage. Only 24 V supply system shall be used for hand lamps etc. while working inside metallic tanks or conducting vessels.

25. Power cable Joints shall be avoided; Unavoidable Joints shall be connected through Cable connector

3. OPERATION & MAINTENANCE

1. All persons, who work with electrical installation/equipment, shall be aware of the electrical hazards, use to protective devices and safe operational procedures. They shall be given training in firefighting, first aid and artificial resuscitation techniques.

2. The CONTRACTOR shall instruct the workers in the proper procedure, specify and enforce the use of necessary protective equipment such as adequately insulated pliers, screw drivers, fuse pulleys, testing lamps and similar hand tools. Only wooden ladders shall be used to reach the heights in electrical work.

3. No material or earthwork shall be allowed to be dumped below or in the vicinity of the bare overhead line conductors.

4. Before any maintenance work is commenced on electrical installations/equipment, the circuits shall be de-energized and ascertained to be dead by positive test with an approved voltage-testing device. Switches shall be tagged or the fuse holders withdrawn before starting the work. Adequate precautions shall be taken in two important aspects viz.

    c. That there shall be no danger from any adjacent live parts and

    d. That there shall be no chances of re-energizing of the equipment on which the persons are working.

5. While working on or near a circuit, whenever possible the use of one hand may be practiced even though the circuit is supposed to be dead. The other hand may preferably be kept in pocket.

6. When it is necessary to touch electrical equipment (for example when checking for overload of motors) back of the hand may be used. Thus, if accidental shock were to cause muscular contractions, one would not 'freeze' to the conductor.

7. Operations of electrical equipment shall be avoided which standing on wet floor or when hands are wet.

8. Before blown fuses are replaced, the circuit shall be locked out and an investigation shall be made for the cause of the short circuit or overload.

9. When two persons are working within reach of each other, they shall never work on difference phases of the supply.

10. When structural repairs, modification or painting work are to be undertaken, appropriate measures shall be taken for the protection of persons whose work may bring them into the priority of live equipment/circuit.

11. It shall be ensured that the insulation and wire size of extension cords are adequate for the voltage and current to be carried.

12. While tapping electricity from the socket, plug top must be used. It shall be ensured that no extension boards are over loaded while tapping. Only standard three pin plugs shall be used for tapping electricity. Broken sockets/plugs shall be replaced immediately with good ones. Only joints free cables shall be used for connecting equipment/ apparatus.
13. Floors shall be kept free from tailing electrical cables to avoid tripping hazard.
14. Power supply to the entire machines and lighting fixture shall be switched off when not in use.
15. Temporary electrical connections shall be removed as soon as the stipulated work is over. After completion of the works, the CONTRACTOR shall dismantle the distribution boards and the other facilities erected at site.
16. Unauthorized/excess tapping of power from distribution boards under the control of the main contractor or client without prior permission is prohibited at all circumstances.
17. No flammable materials shall be stored in any working area near & near the switchboards where there is a possibility of spark or fire
18. "MEN ON LINE" "DO NOT SWITCH ON" "DANGER" OR "CAUTION" boards as applicable shall be used during maintenance works on the electrical equipment.

4. **PORTABLE ELECTRICAL EQUIPMENT**

5. Portable electrical equipment shall be regularly examined, tested and maintained to ensure that the equipment and its leads are in good order. Register shall be maintained for inspection, recording the testing dates and results of the equipment.
6. All portable appliances (with metal body) shall be provided with three-core cable and three-pin plug. The third pin of the plug shall invariably be earthen. It shall be ensured that the metal part of the equipment shall be effectively earthen.
7. All connections to portable equipment or machines from the panel/distribution board/extension board shall be taken using 3 core double insulated PVC flexible copper wire in one length. No joints shall be allowed in this flexible wire. In case single length of wire is not sufficient for a particular location then the supply can be tapped by providing another extension board or using industrial cable connectors comprising of switch and socket.
8. Flexible cables for portable lamps, tools and apparatus shall be regularly examined, tested periodically and maintained to ensure safety.

55) SAFETY CODES IN CONSTRUCTION INDUSTRY

35.1. EXCAVATION (IS-3764)

xiii. Location and protection of underground utility–viz. Water main, cables.
xiv. Barricade – openings.
xv. Lamp at night – warning signs.
xvi. Excavated material – 5 ft. Away
xvii. Shoring for sides – avoid sides from collapsing.
xviii. Avoid vertical cuts.

xix. Water controlled.
xx. Equipment / vehicles at safe distance.
xxi. Ensure stability of other structures nearby.
xxii. Do not take gas cylinders inside trenches.
xxiii. Trenches greater than 4 ft. Depth should be provided with shoring – ladder for each 15m for hazardous area and each 30m for low hazardous area.
xxiv. Excavated portion.

35.2. CEMENT WORKS

Cement dermatitis, burns-use gumboots, gloves, and respirators.

35.3. SCAFFOLDING (IS 4014(PART II))

xiv. Clear the area of unwanted materials.
xv. Erection under proper supervision.
xvi. Ensure ropes & cables are in good condition.
xvii. Ensure that all structural members and all connections adequate.
xviii. Supports strong – adequate cross bracing.
xix. Ensure ground is safe and provide proper foot hold.
xx. Keep ladder/working area free of grease/oil.
xxi. Provide guardrails & toe board.
xxii. Wear safety belt, helmet while working on scaffolding.
xxiii. Do not use the scaffolding for more than 15 days without rechecking.
xxiv. Frequent inspection.
xxv. Excavation is not permitted near base of scaffolding.
xxvi. Use of wooden scaffolding is not permitted

35.4. LADDERS (IS 3696 (PART II)

i. Proper inspection – rungs/steps are not spliced.
ii. Properly secured – top & bottom.
iii. Side rails on fixed ladders to extend above top landing.
iv. Build up ladders of sound material.
v. Rungs not to exceed 12 inches.
vi. Step ladders fully open during use.
vii. Metal ladders – prohibited near electrical lines. Proper maintenance and storage after use.
viii. Rails of the ladder placed at the roof should extend minimum 1.05m (Vertical height from work Platform to Top of the Ladder).

35.5. BARRICADES

viii. Floor openings covered/barricaded properly.
ix. Road ways/sidewalks protected.
x. Traffic controlled.
xi. walkways
xii. Excavation of trenches
xiii. Hoisting/Lifting scene
xiv. Hazardous areas like radiation test area, Demolishing area, Deshuttering area, Electrical hazardous area, Painting Yard

35.6. CONCRETE CONSTRUCTION

viii. Forms properly installed and braced.
ix. Proper curing period & procedure.
x. Mixing & transporting equipment properly supported/routed.
xi. Adequate runways.
xii. Protection from cement dust.
xiii. Hard hats – safety shoes, skin covering.
xiv. Nails to be removed from shuttering material.

35.7. MASONRY

iv. Proper scaffolding.
v. Dust protection.
vi. Safe hoisting equipment.

35.8. HOISTS, CRANES, DERRICKS

xviii. Inspect – cables, slings, chains, hooks, eyes
xix. Equipment stability/supports.
xx. Out riggers used if required.
xxi. Power lines removed/inactivated (cranes).
xxii. Signals understood and observed.
xxiii. Experienced operators; medical fitness , color vision test certificate for operators & signal men
xxiv. All equipment properly lubricated/maintained.
xxv. Protective head gears.
xxvi. If person climbs on derrick installed for lifting material, special precautions should be taken.
xxvii. Periodical test/inspection of hoists and cranes.
xxviii. Mark “safe working loads” on hoists/cranes.
xxix. Cranes to be operated by certified operator.
xxx. Ensure that slings are vertical.
xxxi. Do not drag chains, slings, hooks and load over the floor.
xxxii. Know the load before lifting.
xxxiii. Barricade around the area to avoid labour movement close to the area
xxxiv. Guide ropes are mandatory while lifting materials with different cranes

35.9. CONVEYORS:

v. Proper inspection & maintenance.
vi. Screens and other protection.
vii. Adequate inspection and maintenance of ladders, light
viii. Emergency Stop Pull Cord

35.10. HEAVY EQUIPMENT & TRUCKS

xiv. Qualified operators with license.
xv. Vehicle laws & regulations to be observed.
xvi. Check – breaks, lights, warning devices and wheels.
xvii. Weight limits – load sizes controlled.
xviii. Haul loads well maintained – properly laid.
xix. Personnel not carried in unsafe manner.
xx. Protection when equipment not used.
xxi. Planned inspection – maintenance.
xxii. Adequate equipment records

xxiii. Proper oil, fuel, lubricants used.
xxiv. Reverse horn to be fixed for vehicles being used in the site.
xxv. Speed limit signage to be installed.
xxvi. Fitness certificate of the vehicles

35.11. HOUSE KEEPING & SANITATION

ix. Proper and organized material stacking
x. General neatness in working area.
xi. Regular disposal of waste/trash.
xii. Pathways and walkways clear.
xiii. Adequate lighting.
xiv. Sanitary facilities – clean.
xv. Adequate drinking water.
xvi. Scrap yard to be marked

35.12. EMERGENCY PROCEDURE

x. First aid station – properly manned.
xi. First aid boxes – with First aid materials
xii. Nearest Hospital tie up for emergency treatment
xiii. All Injuries reported promptly.
xiv. Adequate emergency evacuation facilities.
xv. Emergency assembly point /Signage / illuminated access
xvi. Emergency evacuation procedure & Emergency Response Plan
xvii. Training for emergency response teams periodically
xviii. Emergency siren , Mega Phone , Whistle , Hand Baton , Reflective Jacket

35.13. WELDING – CUTTING

xvi. Qualified & experienced technicians
xvii. Proper screen, shield, goggles, gloves, clothing, equipment.
xviii. Electrical equipment – grounded – inspected.
xix. Power cables protected & in good condition.
xx. Fire extinguishers – available.
xxi. Inspection of fire hazards.
xxii. Flammable materials – protected (40 feet away).
xxiii. Gas cylinders – upright & chained.
xxiv. Gas line, torch in good condition.
xxv. Trolleys at the time of moving cylinders in the working areas
xxvi. Cylinder stand for storage in Cylinder Yard
xxvii. Proper covering of mezzanine holes or barricades for cut portion.
xxviii. Barricades/notices below mezzanine being cut – work permit
xxix. Helper also to use goggles.
xxx. Flash Back arresters to be provided at both end of cylinders and torch

35.14. FLAMMABLE GASES – LIQUIDS

xi. "No smoking" signboard.
xii. Containers clearly identified/marked.
xiii. Proper storage practices.

xiv. Proper storage temperature – protection.
xv. Fire hazards to be checked.
xvi. Proper & adequate fire extinguishers.
xvii. Neat storage area – clear passages.
xviii. Material firmly stacked – not too high.
xix. Entry restricted.
xx. Store in separate enclosed area.

35.15. HANDLING AND STORAGE

ix. Proper number for operation.
x. Person picking up leads correctly.
xi. Materials protected from heat/moisture.
xii. Protection from falling into hoppers/bins.
xiii. Dust protection observed.
xiv. Extinguishers/fire protection available.
xv. Traffic routing and control.
xvi. Safety precautions in line with MSDS- Material safety data sheet

35.16. POWER TOOLS

xiii. Good housekeeping where used.
xiv. Tools – cords – earthing – in good condition.
xv. Proper instructions for use.
xvi. Proper mechanical safe guards.
xvii. Tools nearly stored when not in use.
xviii. Right tool for job.
xix. Proper wiring.
xx. Proper training to operator.
xxi. Proper supervision.
xxii. Use of safety appliances - goggles - face shield.
xxiii. Flying hazards checked up.
xxiv. Work shall be carried out after obtaining work permit

35.17. EDUCATING LABOUR:

vii. Safety Posters/boards to be installed at site at different locations in local languages
viii. Safety Officer to conduct safety meets periodically and holds pep talks with workmen.
ix. Mock drills & fire drills
x. Awards for motivating Labours
xi. Tool box talks,
xii. specialized training for high risk activities

56) SCHEDULE OF FINES

Schedule of Charges to Contractors who are in breach of the employer's Site Safety, Site Safety Cycle and Environmental Rules and Regulations

Noncompliance of content of this document content may result in penalty for first violation, stoppage of work, removal of supervisor or member from the work premises for second violation.

1	Smoking in an unauthorized area and/or consumption of alcohol and/or use of illegal substances.	Rs 500.00
2	Burning of waste or smoldering of combustible materials on site other than for heat treatment processes required for the execution of the Sub Contract works.	Rs 2,000.00
3	Failure to wear personal protective equipment (P.P.E.) eg. Safety helmets, safety boots, goggles etc. respirator, ear plugs, safety belts which shall include failure to anchor belt to a secure structure. Where any site operation requires the use of PPE then all workmen must use the required PPE eg. grinding, welding, burning, unloading hazardous materials etc.	1) Rs.500.00 per worker when lack of enforcement of the usage of P.P.E. by the Sub Contractor is observed by the employer. OR 2) Rs.1, 000.00 where issuance of the required P.P.E. by the Sub Contractor equipment is not carried out.
4	Failure to attend general safety induction course conducted by the employer / the employer.	1) Rs.300.00 per worker for not attending the course; and 2) Workers to attend course within 2 working days or be dismissed.
5.	Failure to attend a notified site safety meeting.	Rs 2,000.00
6.	Failure to submit, within the specified time to the employer, safety supervisor reports which shall include other relevant statutory reports made under the F&IUO Cap. 59, tool box briefing records, weekly labour return, issuance of personal protective equipment records, safety data sheets of toxic and harmful materials and others related certificates.	Rs 2,000.00
7.	Failure to submit a written report for an accident and/or other dangerous occurrence, to the employer within 24 hours of its occurrence.	Rs 2,000.00
8.	Failure to carry out within the specified time the necessary improvement action against any notified safety violation.	Rs 5,000.00
9.	Damaged to or misuse of the employer's property.	1) Rs.2,000.00; and 2) In addition the Sub Contractor to pay for the cost of items damaged.
10.	Failure to maintain work area, facility storage and preparation yard, office premises and workers changing and	1) Rs 2,000.00; and

	rest area in a clean and orderly state and free from health and fire hazards.	2) In addition the Sub Contractor shall clean up the disorderly and untidy areas within 3 days.
11.	Obstruction of passageways, entrances, doorways, stairs, access to firefighting equipment etc. and/or the erection unsafe accesses and crossings.	Rs 2,000.00
12.	Use of equipment that has not been examined by an approved person as required under the Factories and Industries Undertaking Ordinance and its related regulations.	Rs 5,000.00
13.	Using defective or uncertified slings for lifting operations.	Rs 5,000.00
14.	Executing unsafe hoisting of materials and include unsafe use of lifting appliances.	Rs 5,000.00
15.	Erecting and/or using unsafe or unstable, scaffolds, working platforms and temporary structures.	Rs 5,000.00
16.	Failure to provide and use proper working platforms and safe means of access to the work place, where work is required to be carried beyond a person's normal reach.	1) 5,000.00; and 2) The worker to be immediately banned from the site if an individual receives 2 violations.
17.	Allowing workers to occupy or work on unguarded elevated platforms, floor edges and without adopting adequate safety measures against the risk of a person falling from height.	Rs 7,000.00
18.	Not providing safety barricades/barriers to hazardous floor edge openings, gaps and shafts.	Rs 7,000.00
19.	Rendering scaffold or working platform unsafe by tampering/alternation.	1) Rs 5,000.00 and 2) All rectification costs to be borne by the subcontractor.
20.	Placing of heavy items unsafely on scaffold or working platform.	Rs 2,000.00
21.	Throwing or allowing objects to drop from heights.	1) Rs 7,000.00 and 2) Workers to be immediately banned from the site.
22.	Stacking or leaving materials including work in progress articles and tools in unstable condition and or along floor edges such they are likely to endanger workers.	Rs 7,000.00
23.	Failure to effectively cordon off guard and warn other workers from entering into the danger areas when they are likely to be affected by falling materials from the Sub Contract Work.	Rs 7,000.00
24.	Violating the permit to work system for example, but not limited to:- d) Look out procedures e) Confined space entry permits f) Other hazardous work Eg. hot work (in the event of failure to comply)	1) Rs 5,000.00; and 2) Workers to be immediately banned from the site.

25.	Dismantling and rendering any safety guards or protective features of any part of a machine or any part of a building structure to the extent that such guards and protective features are not operational or are incapable of providing the necessary protection for its design and purpose.	Rs 7,000.00
26.	Adopting unsafe tapping, connections, and termination of electrical lines and including the use of defective electrical fittings, power cables and electrical tools. Allowing cables / equipment to be submerged in water.	Rs 2,000.00
27.	Using any defective or unsafe equipment.	Rs 2,000.00
28.	Unauthorized use of fire equipment provided for emergency purposes.	Rs 2,000.00
29.	Failure to comply with an order issued by the employer's Construction manager, site agent, site safety supervisor safety officer and environmental officers in regard to safety/environmental matters.	Upto Rs 7,000.00
30.	Threatening safety/personnel misbehavior, fighting or intentional causing hurt to others.	1) Rs 7,000.00 & to be blacklisted and 2) Worker to be banned from the site and a report will be made to the police.
31.	Other safety/environmental violations which are in breach of safety or environmental ordinances and regulations.	Rs 1,000.00
32.	Failure to wear safety harness and anchor to a secure structure, whilst working on a gondola.	Rs 7,000.00 and worker to be banned from site immediately.
33.	Tampering with the safety features of a gondola / winches ./ scaffolding	Rs 7,000.00 and worker to be banned from site immediately.
34.	Access and egress to gondola other than at a designated landing.	Rs7,000.00 and worker to be banned from site immediately.
35.	Failure to provide valid certificates for lifting appliances and accessories including any lifting appliances / accessories on vehicles delivering goods to the site, or for plant being used on site.	Rs5,000.00
36.	Failure to provide voltage-reducing devices on welding machines.	Rs 2,000.00
37.	Obstruction of passageways, entrances, doorways, stairs, access to firefighting equipment etc. and/or the erection unsafe accesses and crossings.	Rs 2,000.00
38.	Failure to attend the Morning Site Assembly Meetings.	Rs 500.00
39.	Failure to tidy up the Sub contract works after each day work	Rs1,200.00
40.	Failure to attend the Site Safety Walk	Rs 1,000.00
41.	Failure to provide Site Safety Supervisor/Representative as per the requirement of the latest ordinance.	Rs 2,000.00
42.	Failure to use stairs in lieu of lifts	Rs 500 per event

T FOR STATUTORY COMPLIANCE REGISTERS
TO BE MAINTAINED BY A CONTRACTOR

Note:

6. ***The following are the mandatory list of registers that need to be maintained as per the statutory requirements (the items may vary as per Rules of the State in which project is being executed)***
7. ***The registers need to be maintained only for the employees deployed at the site***
8. ***One set of photocopies of all the documents to be submitted at the time of audit***
9. ***The following documents to be maintained for the period commencement of contract to completion of contract***

20. Contract License
21. Form VI-A commencement of Contract intimation to the department
22. Muster Roll
23. Wage Register
24. Register of overtime
25. Register of Advance
26. Register of Fine
27. Register of Workman Employed by the Contractor
28. Employment Card
29. In & out timings register
30. PF registration copy
31. PF Challans
32. PF contribution details for the employees deployed at the site (from PF web portal)
33. ESIC Challans
34. ESIC Temp card copy of for the employees deployed at the site
35. ESIC Contribution details for the employees deployed at the site (from ESIC web portal)
36. Vendors monthly compliance undertaking (as per the excel sheet attached)

VENDOR MONTHLY COMPLAINCES FORMAT

Name & Address of the Vendor	Name of the Principal Employer & Address	Month	
		ESIC.Reg. No.	
		EPF . Reg No.	
:		Professional Tax. Re	

SL No.	Name of the employee	Father's Name	Employee's PF Number	Employee's ESI Number	No.of days worked during the month	Basic wage	DA	HRA &Other Allowance	Over time Hour	Overtime wages	Gross	DeductionOf EPF	Deduction ESIC	Deduction PT	Deduction LWF	NET Amount

We hereby Certify that the above mentioned employees PF/ESIC has been remitted with the respective department against challan attached for the Month of..... .. We hereby affirm that we will be liable to pay any financial liabilities in case of such liabilities arise in future in regard to the documents submitted by us.

37. Labor Welfare Fund contribution at Haryana
38. Cess payment Receipt (As per the BOCW registration obtained by Client)

Any other document/Register as may be required per the prevailing Statutory/Legal requirements of the State

10. <u>The attached documents to be maintained monthly labour compliance audit will be conducted. Final bill payment will be released based on labour compliance audit clearance.</u>

AUDIT CHECKLIST						
SITE NAME :		DATE OF AUDIT				
LOCATION:		PLACE OF AUDIT				
NAME OF CONTRACTOR		NAME OF AUDITOR				
ADDRESS:		No of Employees				
		Nature OF Work				
NAME OF REPRESENTATIVE						
CONTACT NUMBER/E-Mail		Contract Period				

Sl No	Act	Compliance check	Responsible	Compliant/Non Compliant	April	
1	BOCW Act	Whether Registration under BOCW Act has been obtained				
2	BOCW Act	Whether Non-Compliantttice of Commencement/completion has been filed				
3	BOCW Act	Whether Cess has been paid				
4	BOCW Act	Whether Annual Return has been filed				
5	BOCW Act	Whether Certificate of Registration has been displayed at site				
6	BOCW Act	Whether Abstract has been displayed at site				
7	BOCW Act	Whether information related to Working Hours, Rate of Wages, Rest Day, Payment date has been displayed at site				
8	BOCW Act	Whether Name of Inspector with Address has been displayed.				
9	BOCW Act	Whether place and time of disbursement of wages has been displayed at site				
10	BOCW Act	Whether overtime has been paid to employees @ double rate of wages				
11	BOCW Act	Whether any employee worked more than 9 hours on any day or more than 48 hours in a week				
12	BOCW Act	Whether interval for rest has been provided to employees on every five hours of working.				
13	BOCW Act	Whether any employee's spread hours are more than 12 hours in a day (including interval for rest)				
14	BOCW / CLRA / MW / PW Act	Whether "Register of Building workers employed by the Employer" has been maintained.				
15	BOCW / CLRA / MW / PW Act	Whether Muster Roll has been maintained				
16	BOCW / CLRA / MW / PW Act	Whether Rigister Of Wages has been maintained				
17	BOCW / CLRA / MW / PW Act	Whether Form Of Register Of Wages-Cum-Muster-Roll has been maintained				
18	BOCW / CLRA / MW / PW Act	Whether Register Of Deductions For Damages Or Loss has been maintained				
19	BOCW / CLRA / MW / PW Act	Whether Register Of Fines has been maintained				
20	BOCW / CLRA / MW / PW Act	Whether Register Of Advances has been maintained				
21	BOCW / CLRA / MW / PW Act	Whether Register Of Overtime has been maintained				
22	BOCW / CLRA / MW / PW Act	Whether Wage Book has been maintained				
23	BOCW / CLRA / MW / PW Act	Whether Service Certificate has been issued to workers				
24	BOCW / CLRA / MW / PW Act	Whether Identity Card Issued To Building Workers				
25	BOCW Act	Whether any accident or dangerous occurred at site				
26	BOCW Act	Whether Register of accident and dangerous occurrence has been maintained				
27	BOCW Act	Whether rest day provided on every six days of working				
28	BOCW Act	In case employee worked on rest day, compensatory off has been provided within 5 days and given overtime for working on rest day.				
29	BOCW Act	In case any worker completed more than 90 days, whether same worker has been registered as beneficiary				
30	BOCW Act	Whether employee and employer contribution has been paid to department on monthly basis				
31	BOCW Act	whether monthly return related to contribution has been filed with department				
32	BOCW Act	Whether Register related to Beneficiary workers has been maintained				
33	CLRA Act	Whether CLRA Applicable (if project period is more than 90 days)				
34	CLRA Act	Labour License and Validity				
35	CLRA Act	Form-VI A for Commincement and Complitions				
36	Minimum Wages Act	Whether applicable minimum wages has been paid to workers				
37	ESIC Act	Whether ESIC is applicable on Establishment				
38	ESIC Act	If applicable then whether ESIC has been deducted from employees salary as per provisions				
39	ESIC Act	Whether ESIC challan has been generated and ECR is matching with deduction as per Wage Register				
40	ESIC Act	Whether ESIC Contribution has been paid on time.				
41	ESIC Act	Whether ESIC contribution has been paid in the local ESIC sub Code or mapped to another location.				
42	PF ACT	Whether PF is applicable on Establishment				
43	PF ACT	Whether company deposit PF with RPFC or have own PF Trust.				
44	PF ACT	If applicable then whether PF has been deducted from employees salary as per provisions				
45	PF ACT	Whether PF challan has been generated and ECR is matching with deduction as per Wage Register				
46	PF ACT	Whether PF Contribution has been paid on time.				
47	Labour Welfare Fund Act	If applicable, whether LWF has been deducted as per applicable slabs				
48	Labour Welfare Fund Act	Whether LWF has been paid on time.				
49	Professional Tax Act	If completed then whether PT has been deducted and paid				
50	Professional Tax Act	If applicable Monthly / Yearly Return				
51	Equal Remuneration Act	Whether wages has been paid to male and female employees at the same rate and there is no difference based on the gender.				
52	Child Labour Act	Whether employed any child labour who is below 15 years of age.				
53	Child Labour Act	Whether employed any young person who has completed 15 years of age but below 18 years of age				
54	Inter State Migrant Labour Act	Whether five or more inter state migrant workers employed				
55	Inter State Migrant Labour Act	Whether Contractor has applied for Licence is this Act is applicable				
56	Inter State Migrant Labour Act	Whether Pass book has been issued to workers as per provision of this Act (If applicable)				
57	Inter State Migrant Labour Act	Whether periodic return has been filed under this Act				
58	Employee Compensation Act	Whether employees who are not covered under ESIC Act, are covered under employee compensation policy.				